

41

Animal Nutrition

KEY CONCEPTS

- 41.1** An animal's diet must supply chemical energy, organic molecules, and essential nutrients
- 41.2** The main stages of food processing are ingestion, digestion, absorption, and elimination
- 41.3** Organs specialized for sequential stages of food processing form the mammalian digestive system
- 41.4** Evolutionary adaptations of vertebrate digestive systems correlate with diet
- 41.5** Feedback circuits regulate digestion, energy storage, and appetite

▲ **Figure 41.1** How does a crab help an otter make fur?

The Need to Feed

Dinnertime has arrived for the sea otter in **Figure 41.1** (and for the crab, though in quite a different sense). The muscles and other organs of the crab will be chewed into pieces, broken down by acid and enzymes in the otter's digestive system, and finally absorbed as small molecules into the body of the otter. Such a process is what is meant by animal **nutrition**: food being taken in, taken apart, and taken up.

Although dining on fish, crabs, urchins, and abalone is the sea otter's specialty, all animals eat other organisms—dead or alive, piecemeal or whole. Unlike plants, animals must consume food for both energy and the organic molecules used to assemble new molecules, cells, and tissues. Despite this shared need, animals have diverse diets. **Herbivores**, such as cattle, sea slugs, and caterpillars, dine mainly on plants or algae. **Carnivores**, such as sea otters, hawks, and spiders, mostly eat other animals. Rats and other **omnivores** (from the Latin *omnis*, all) don't in fact eat everything, but they do regularly consume animals as well as plants or algae. We humans are typically omnivores, as are cockroaches and crows.

The terms *herbivore*, *carnivore*, and *omnivore* represent the kinds of food an animal usually eats. Keep in mind, however, that most animals are opportunistic feeders, eating foods outside their standard diet when their usual foods aren't available.

For example, deer are herbivores, but in addition to feeding on grass and other plants, they occasionally eat insects, worms, or bird eggs. Note as well that microorganisms are an unavoidable “supplement” in every animal’s diet.

Animals must eat. But to survive and reproduce, they must also balance their consumption, storage, and use of food. Sea otters, for example, support a high rate of metabolism by eating up to 25% of their body mass each day. Eating too little food, too much food, or the wrong mixture of foods can endanger an animal’s health. In this chapter, we’ll survey the nutritional requirements of animals, explore diverse evolutionary adaptations for obtaining and processing food, and investigate the regulation of energy intake and expenditure.

CONCEPT 41.1

An animal’s diet must supply chemical energy, organic molecules, and essential nutrients

Overall, an adequate diet must satisfy three nutritional needs: chemical energy for cellular processes, organic building blocks for macromolecules, and essential nutrients.

The activities of cells, tissues, organs, and whole animals depend on sources of chemical energy in the diet. This energy is used to produce ATP, which powers processes ranging from DNA replication and cell division to vision and flight. To meet the continuous requirement for ATP, animals ingest and digest nutrients, including carbohydrates,

proteins, and lipids, for use in cellular respiration and energy storage.

In addition to providing fuel for ATP production, an animal’s diet must supply the raw materials needed for biosynthesis. To build the complex molecules it needs to grow, maintain itself, and reproduce, an animal must obtain two types of organic precursors from its food. Animals need a source of organic carbon (such as sugar) and a source of organic nitrogen (such as protein). Starting with these materials, animals can construct a great variety of organic molecules.

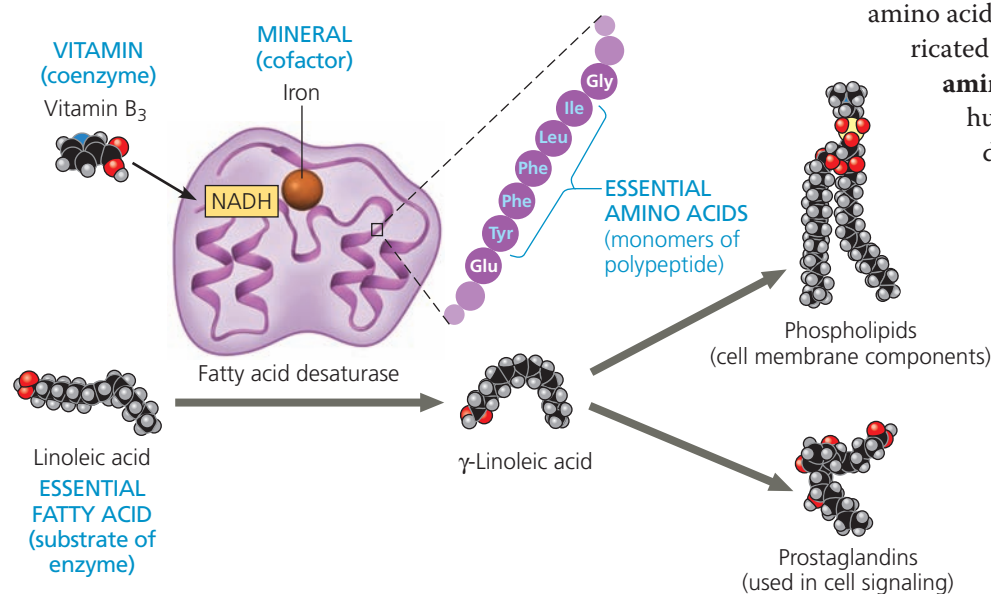
Essential Nutrients

Some cellular processes require materials that an animal cannot assemble from simpler organic precursors. These materials—preassembled organic molecules and minerals—are called **essential nutrients**. Obtained from an animal’s diet, essential nutrients include essential amino acids and fatty acids, vitamins, and minerals. Essential nutrients have key functions in cells, including serving as substrates of enzymes, as coenzymes, and as cofactors in biosynthetic reactions (**Figure 41.2**). Needs for particular nutrients vary among species. For instance, ascorbic acid (vitamin C) is an essential nutrient for humans and other primates, as well as guinea pigs, but not for many other animals.

Essential Amino Acids

Animals require 20 amino acids to make proteins (see **Figure 5.14**). Most animal species have the enzymes to synthesize about half of these amino acids, as long as their diet includes sulfur and organic nitrogen. The remaining amino acids must be obtained from food in prefabricated form and are therefore called **essential amino acids**. Many animals, including adult humans, require eight amino acids in their diet: isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan, and valine. (Human infants also need a ninth, histidine.)

The proteins in animal products such as meat, eggs, and cheese are “complete,” which means that they provide all the essential amino acids in their proper proportions. In contrast, most plant proteins are “incomplete,” being deficient in one or more essential amino acids. Corn (maize), for example, is deficient in tryptophan and lysine, whereas beans are lacking in methionine. However, vegetarians can easily obtain all of the essential amino acids by eating a varied diet of plant proteins.



▲ **Figure 41.2 Roles of essential nutrients.** Linoleic acid is converted by the enzyme fatty acid desaturase to γ -linolenic acid, a precursor for phospholipids and prostaglandins. This biosynthetic reaction illustrates common functions of the four classes of essential nutrients, labeled in blue. Note that nearly every enzyme or other protein in animal bodies contains some essential amino acids, as indicated in the partial sequence shown for fatty acid desaturase.

Essential Fatty Acids

Animals require fatty acids to synthesize a variety of cellular components, including membrane phospholipids, signaling molecules, and storage fats. Although animals can synthesize many fatty acids, they lack the enzymes to form the double bonds found in certain required fatty acids. Instead, these molecules must be obtained from the diet and are considered **essential fatty acids**. In mammals, they include linoleic acid (see Figure 41.2). Because seeds, grains, and vegetables generally furnish ample quantities of essential fatty acids, deficiencies in this class of nutrients are rare.

Vitamins

As Albert Szent-Györgyi, the discoverer of vitamin C, once quipped, “A vitamin is a substance that makes you ill if you *don’t* eat it.” **Vitamins** are organic molecules that are required in the diet in very small amounts. They have diverse functions. Vitamin B₂, for example, is converted in the body to FAD, a coenzyme used in many metabolic processes, including cellular respiration (see Figure 9.12). For humans, 13 vitamins have been identified. Depending on the vitamin, the required amount ranges from 0.01 to 100 mg per day.

Vitamins are classified as water-soluble or fat-soluble (**Table 41.1**). B vitamins, which generally act as coenzymes, are water-soluble. So is vitamin C, which is required for the production of connective tissue. Fat-soluble vitamins include vitamin A, which is incorporated into visual pigments of the eye, and vitamin D, which aids in calcium absorption and bone formation. The dietary requirement for vitamin D is variable in humans because we can synthesize it from other molecules when our skin is exposed to sunlight.

For people with imbalanced diets, taking vitamin supplements that provide recommended daily levels is certainly reasonable. It is far less clear that massive doses of vitamins confer any health benefits or are even safe. Moderate overdoses of water-soluble vitamins are probably harmless because excesses are excreted in urine. However, excesses of fat-soluble vitamins are deposited in body fat, so overconsumption may cause them to accumulate to toxic levels.

Minerals

Dietary **minerals** are inorganic nutrients, such as iron and sulfur, that are usually required in small amounts—from less than 1 mg to about 2,500 mg per day. As shown in **Table 41.2**,

Table 41.1 Vitamin Requirements of Humans			
Vitamin	Major Dietary Sources	Major Functions in the Body	Symptoms of Deficiency
Water-Soluble Vitamins			
B ₁ (thiamine)	Pork, legumes, peanuts, whole grains	Coenzyme used in removing CO ₂ from organic compounds	Beriberi (tingling, poor coordination, reduced heart function)
B ₂ (riboflavin)	Dairy products, meats, enriched grains, vegetables	Component of coenzymes FAD and FMN	Skin lesions, such as cracks at corners of mouth
B ₃ (niacin)	Nuts, meats, grains	Component of coenzymes NAD ⁺ and NADP ⁺	Skin and gastrointestinal lesions, delusions, confusion
B ₅ (pantothenic acid)	Meats, dairy products, whole grains, fruits, vegetables	Component of coenzyme A	Fatigue, numbness, tingling of hands and feet
B ₆ (pyridoxine)	Meats, vegetables, whole grains	Coenzyme used in amino acid metabolism	Irritability, convulsions, muscular twitching, anemia
B ₇ (biotin)	Legumes, other vegetables, meats	Coenzyme in synthesis of fat, glycogen, and amino acids	Scaly skin inflammation, neuromuscular disorders
B ₉ (folic acid)	Green vegetables, oranges, nuts, legumes, whole grains	Coenzyme in nucleic acid and amino acid metabolism	Anemia, birth defects
B ₁₂ (cobalamin)	Meats, eggs, dairy products	Production of nucleic acids and red blood cells	Anemia, numbness, loss of balance
C (ascorbic acid)	Citrus fruits, broccoli, tomatoes	Used in collagen synthesis; antioxidant	Scurvy (degeneration of skin and teeth), delayed wound healing
Fat-Soluble Vitamins			
A (retinol)	Dark green and orange vegetables and fruits, dairy products	Component of visual pigments; maintenance of epithelial tissues	Blindness, skin disorders, impaired immunity
D	Dairy products, egg yolk	Aids in absorption and use of calcium and phosphorus	Rickets (bone deformities) in children, bone softening in adults
E (tocopherol)	Vegetable oils, nuts, seeds	Antioxidant; helps prevent damage to cell membranes	Nervous system degeneration
K (phylloquinone)	Green vegetables, tea; also made by colon bacteria	Important in blood clotting	Defective blood clotting

minerals have diverse functions in animal physiology. Some are assembled into the structure of proteins; iron, for example, is incorporated into the oxygen carrier hemoglobin as well as some enzymes (see Figure 41.2). In contrast, sodium, potassium, and chloride are important in the functioning of nerves and muscles and in maintaining osmotic balance between cells and the surrounding body fluid. In vertebrates, the mineral iodine is incorporated into thyroid hormone, which regulates metabolic rate. Vertebrates also require relatively large quantities of calcium and phosphorus for building and maintaining bone.

Ingesting large amounts of some minerals can upset homeostatic balance and impair health. For example, excess salt (sodium chloride) can contribute to high blood pressure. This is a particular problem in the United States, where the typical person consumes enough salt to provide about 20 times the required amount of sodium. Packaged (prepared) foods often contain large amounts of sodium chloride, even if they do not taste very salty.

Dietary Deficiencies

A diet that lacks one or more essential nutrients or consistently supplies less chemical energy than the body requires results in *malnutrition*, a failure to obtain adequate nutrition. Malnutrition resulting from either type of dietary deficiency can have negative impacts on health and survival and affects one out of four children worldwide.

Deficiencies in Essential Nutrients

Insufficient intake of essential nutrients can cause deformities, disease, and even death. For example, cattle, deer, and other herbivores may develop dangerously fragile bones if they graze on plants growing in soil that lacks phosphorus. In such environments, some grazing animals obtain missing nutrients by consuming concentrated sources of salt or other minerals (**Figure 41.3**).

Similarly, some birds supplement their diet with snail shells, and certain tortoises ingest stones.



▲ **Figure 41.3** Obtaining essential nutrients from an unusual source. A juvenile chamois (*Rupicapra rupicapra*), an herbivore, licks salts from exposed rocks in its alpine habitat. This behavior is common among herbivores that live where soils and plants provide insufficient amounts of minerals, such as sodium, calcium, phosphorus, and iron.

Table 41.2 Mineral Requirements of Humans*				
Mineral	Major Dietary Sources	Major Functions in the Body	Symptoms of Deficiency	
More than 200 mg per day required	Calcium (Ca)	Dairy products, dark green vegetables, legumes	Bone and tooth formation, blood clotting, nerve and muscle function	Impaired growth, loss of bone mass
	Phosphorus (P)	Dairy products, meats, grains	Bone and tooth formation, acid-base balance, nucleotide synthesis	Weakness, loss of minerals from bone, calcium loss
	Sulfur (S)	Proteins from many sources	Component of certain amino acids	Impaired growth, fatigue, swelling
	Potassium (K)	Meats, dairy products, many fruits and vegetables, grains	Acid-base balance, water balance, nerve function	Muscular weakness, paralysis, nausea, heart failure
	Chlorine (Cl)	Table salt	Acid-base balance, formation of gastric juice, nerve function, osmotic balance	Muscle cramps, reduced appetite
	Sodium (Na)	Table salt	Acid-base balance, water balance, nerve function	Muscle cramps, reduced appetite
	Magnesium (Mg)	Whole grains, green leafy vegetables	Enzyme cofactor; ATP bioenergetics	Nervous system disturbances
Iron (Fe)	Meats, eggs, legumes, whole grains, green leafy vegetables	Component of hemoglobin and of electron carriers; enzyme cofactor	Iron-deficiency anemia, weakness, impaired immunity	
Fluorine (F)	Drinking water, tea, seafood	Maintenance of tooth structure	Higher frequency of tooth decay	
Iodine (I)	Seafood, iodized salt	Component of thyroid hormones	Goiter (enlarged thyroid gland)	

*Additional minerals required in trace amounts include cobalt (Co), copper (Cu), manganese (Mn), molybdenum (Mo), selenium (Se), and zinc (Zn). All of these minerals, as well as those in the table, can be harmful in excess.

Like other animals, humans sometimes suffer from diets lacking in essential nutrients. A diet that provides insufficient amounts of one or more essential amino acids causes protein deficiency, the most common type of malnutrition among humans. In children, protein deficiency may arise if their diet shifts from breast milk to foods that contain relatively little protein, such as rice. Such children, if they survive infancy, often have impaired physical and mental development.

In populations subsisting on simple rice diets, individuals are often deficient in vitamin A, which can result in blindness or death. To overcome this problem, scientists have engineered “Golden Rice,” a strain of rice that synthesizes the orange-colored pigment beta-carotene, which the body converts to vitamin A (see Concept 38.3).

Undernutrition

A diet that fails to provide adequate sources of chemical energy results in *undernutrition*. When an animal is undernourished, a series of events unfold: The body uses up stored carbohydrates and fat and then begins breaking down its own proteins for fuel; muscles begin to decrease in size; and the brain may become protein-deficient. If energy intake remains less than energy expenditures, the animal will eventually die. Even if a seriously undernourished animal survives, some of the damage may be irreversible.

Human undernutrition is most common when drought, war, or another crisis severely disrupts the food supply. In sub-Saharan Africa, where the AIDS epidemic has crippled both rural and urban communities, approximately 200 million children and adults cannot obtain enough food.

Sometimes undernutrition occurs within well-fed human populations as a result of eating disorders. For example, anorexia nervosa leads individuals, usually female, to starve themselves compulsively.

Assessing Nutritional Needs

Determining the ideal diet for the human population is an important but difficult problem for scientists. As objects of study, people present many challenges. Unlike laboratory animals, humans are genetically diverse. They also live in settings far more varied than the stable and uniform environment that scientists use to facilitate comparisons in laboratory experiments. Ethical concerns present an additional barrier. For example, it is not acceptable to investigate the nutritional needs of children in a way that might harm a child’s growth or development.

Many insights into human nutrition have come from *epidemiology*, the study of human health and disease at the population level. In the 1970s, for instance, researchers discovered that children born to women of low socioeconomic status were more likely to have neural tube defects, which occur when tissue fails to enclose the developing brain and

▼ Figure 41.4

Inquiry

Can diet influence the frequency of birth defects?

Experiment Richard Smithells, of the University of Leeds, in England, examined the effect of vitamin supplementation on the risk of neural tube defects. Women who had had one or more babies with such a defect were put into two study groups. The experimental group consisted of those who were planning a pregnancy and began taking a multivitamin at least four weeks before attempting conception. The control group, who were not given vitamins, included women who declined them and women who were already pregnant. The numbers of neural tube defects resulting from the pregnancies were recorded for each group.

Results

Group	Number of Infants/Fetuses Studied	Infants/Fetuses with a Neural Tube Defect
Vitamin supplements (experimental group)	141	1
No vitamin supplements (control group)	204	12

Conclusion This controlled study provided evidence that vitamin supplementation protects against neural tube defects, at least after the first pregnancy. Follow-up trials demonstrated that folic acid alone provided an equivalent protective effect.

Source: R. W. Smithells et al., Possible prevention of neural-tube defects by periconceptional vitamin supplementation, *Lancet* 315:339–340 (1980).

Inquiry in Action Read and analyze the original paper in *Inquiry in Action: Interpreting Scientific Papers*.

INTERPRET THE DATA After folic acid supplementation became standard in the U.S., the frequency of neural tube defects dropped to an average of just 1 in 5,000 live births. Propose two explanations why the observed frequency was much higher in the experimental group of the Smithells study.

WHAT IF? Subsequent studies were designed to learn if folic acid supplements prevent neural tube defects during first-time pregnancies. To determine the required number of subjects, what type of additional information did the researchers need?

spinal cord (see Concept 47.2). The English scientist Richard Smithells thought that malnutrition among these women might be responsible. As described in **Figure 41.4**, he found that vitamin supplementation greatly reduced the risk of neural tube defects. In other studies, he obtained evidence that folic acid (vitamin B₉) was the specific vitamin responsible, a finding confirmed by other researchers. Based on this evidence, the United States in 1998 began to require that folic acid be added to enriched grain products used to make bread, cereals, and other foods. Follow-up studies have documented the effectiveness of this program in reducing the frequency of neural tube defects. Thus, at a time when microsurgery and sophisticated diagnostic imaging dominate

the headlines, a simple dietary change such as folic acid supplementation may be among the greatest contributors to human health.

CONCEPT CHECK 41.1

1. All 20 amino acids are needed to make animal proteins. Why aren't they all essential to animal diets?
2. **MAKE CONNECTIONS** Considering the role of enzymes in metabolic reactions (see Concept 8.4), explain why vitamins are required in very small amounts in the diet.
3. **WHAT IF?** If a zoo animal eating ample food shows signs of malnutrition, how might a researcher determine which nutrient is lacking in its diet?

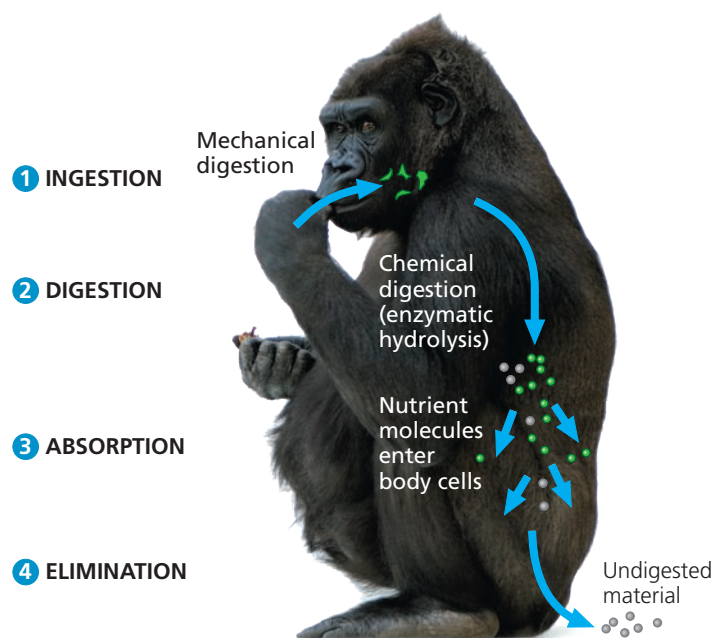
For suggested answers, see Appendix A.

CONCEPT 41.2

The main stages of food processing are ingestion, digestion, absorption, and elimination

In this section, we turn from nutritional requirements to the mechanisms by which animals process food. Food processing can be divided into four distinct stages: ingestion, digestion, absorption, and elimination (**Figure 41.5**).

The first stage, **ingestion**, is the act of eating or feeding. Given the variation in food sources among animal species, it is not surprising that strategies for extracting resources from food also differ widely. **Figure 41.6**, on the next page, surveys and classifies the principal feeding mechanisms



▲ **Figure 41.5** The stages of food processing.

that have evolved in animals. We will focus in this chapter on the shared aspects of food processing, pausing periodically to consider some adaptations to particular diets or environments.

During **digestion**, the second stage of food processing, food is broken down into molecules small enough for the body to absorb. Mechanical digestion, such as chewing, typically precedes chemical digestion. Mechanical digestion breaks food into smaller pieces, increasing the surface area available for chemical processes. Chemical digestion is necessary because animals cannot directly use the proteins, carbohydrates, nucleic acids, fats, and phospholipids in food. One problem is that these molecules are too large to pass through membranes and enter the cells of the animal. In addition, the large molecules in food are not all identical to those the animal needs for its particular tissues and functions. When large molecules in food are broken down into their components, however, the animal can use these smaller molecules to assemble the large molecules it needs. For example, although fruit flies and humans have very different diets, both convert proteins in their food to the same 20 amino acids from which they assemble all of the specific proteins in their bodies.

A cell makes a macromolecule or fat by linking together smaller components; it does so by removing a molecule of water for each new covalent bond formed. Chemical digestion by enzymes reverses this process by breaking bonds through the addition of water (see Figure 5.2). This splitting process is called *enzymatic hydrolysis*. A variety of enzymes catalyze the digestion of large molecules in food. Polysaccharides and disaccharides are split into simple sugars; proteins are broken down into amino acids; and nucleic acids are cleaved into nucleotides and their components. Enzymatic hydrolysis also releases fatty acids and other components from fats and phospholipids. In many animals, such as the gorilla in Figure 41.5, digestion of some materials is accomplished by bacteria living in the digestive system.

The last two stages of food processing occur after the food is digested. In the third stage, **absorption**, the animal's cells take up (absorb) small molecules such as amino acids and simple sugars. **Elimination** completes the process as undigested material passes out of the digestive system.

Digestive Compartments

In our overview of food processing, we have seen that digestive enzymes hydrolyze the same biological materials (such as proteins, fats, and carbohydrates) that make up the bodies of the animals themselves. How, then, are animals able to digest food without digesting their own cells and tissues? The evolutionary adaptation that allows animals to avoid self-digestion is the processing of food within specialized intracellular or extracellular compartments.

Exploring Four Main Feeding Mechanisms of Animals

Filter Feeding



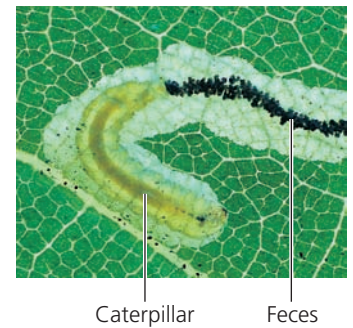
Many aquatic animals are **filter feeders**, which strain small organisms or food particles from the surrounding medium. The humpback whale, shown above, is one example. Attached to the whale's upper jaw are comblike plates called baleen, which remove small invertebrates and fish from enormous volumes of water and sometimes mud. Filter feeding in water is a type of suspension feeding, which also includes removing suspended food particles from the surrounding medium by capture or trapping mechanisms.

Bulk Feeding

Most animals, including humans, are **bulk feeders**, which eat relatively large pieces of food. Their adaptations include tentacles, pincers, claws, venomous fangs, jaws, and teeth that kill their prey or tear off pieces of meat or vegetation. In this amazing scene, a rock python is beginning to ingest a gazelle it has captured and killed. Snakes cannot chew their food into pieces and must swallow

Substrate Feeding

Substrate feeders are animals that live in or on their food source. This leaf miner caterpillar, the larva of a moth, is eating through the soft tissue of an oak leaf, leaving a dark trail of feces in its wake. Other substrate feeders include maggots (fly larvae), which burrow into animal carcasses.



Fluid Feeding

Fluid feeders suck nutrient-rich fluid from a living host. This mosquito has pierced the skin of its human host with hollow, needlelike mouthparts and is consuming a blood meal (colorized SEM). Similarly, aphids are fluid feeders that tap the phloem sap of plants. In contrast to such parasites, some fluid feeders actually benefit their hosts. For example, hummingbirds and bees move pollen between flowers as they fluid-feed on nectar.



it whole—even if the prey is much bigger than the diameter of the snake. They can do so because the lower jaw is loosely hinged to the skull by an elastic ligament that permits the mouth and throat to open very wide. After swallowing its prey, which may take more than an hour, the python will spend two weeks or longer digesting its meal.



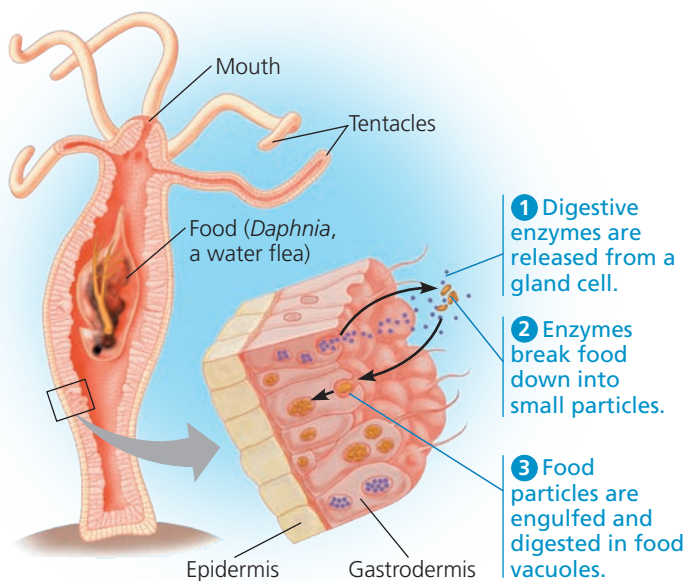
Intracellular Digestion

Food vacuoles—cellular organelles in which hydrolytic enzymes break down food—are the simplest digestive compartments. The hydrolysis of food inside vacuoles, called intracellular digestion, begins after a cell engulfs solid food by phagocytosis or liquid food by pinocytosis (see Figure 7.19). Newly formed food vacuoles fuse with lysosomes, organelles containing hydrolytic enzymes. This fusion of organelles brings food in contact with the enzymes, allowing digestion to occur safely within a compartment enclosed by a protective membrane. A few animals, such as sponges, digest their food entirely by this intracellular mechanism (see Figure 33.4).

Extracellular Digestion

In most animal species, hydrolysis occurs largely by extracellular digestion, the breakdown of food in compartments that are continuous with the outside of the animal's body. Having one or more extracellular compartments for digestion enables an animal to devour much larger pieces of food than can be ingested by phagocytosis.

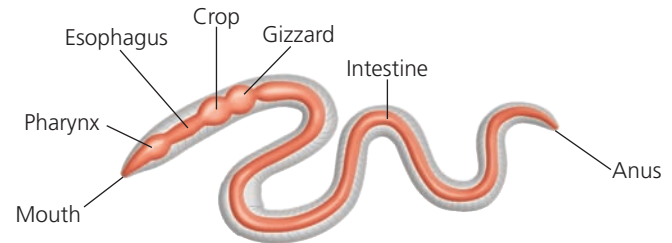
Many animals with relatively simple body plans have a digestive compartment with a single opening (Figure 41.7). This pouch, called a **gastrovascular cavity**, functions in digestion as well as in the distribution of nutrients throughout the body (hence the *vascular* part of the term). The cnidarians called hydras provide a good example of how a gastrovascular cavity works. A carnivore, the hydra uses its tentacles to stuff captured prey through its mouth into its gastrovascular cavity. Specialized gland cells of the hydra's gastrodermis, the tissue layer that lines the cavity, then secrete digestive enzymes that break the soft tissues of the prey into tiny pieces. Other cells of the gastrodermis engulf these



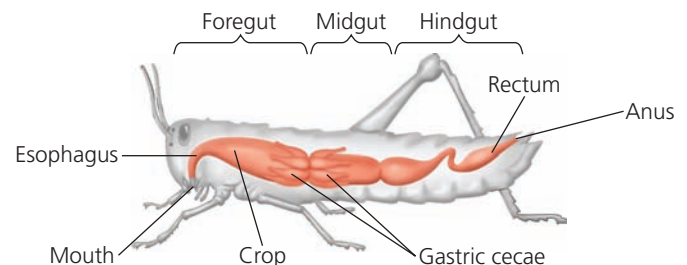
▲ **Figure 41.7 Digestion in a hydra.** Digestion begins in the gastrovascular cavity and is completed intracellularly after small food particles are engulfed by specialized cells of the gastrodermis.

food particles, and most of the hydrolysis of macromolecules occurs intracellularly, as in sponges. After the hydra has digested its meal, undigested materials that remain in its gastrovascular cavity, such as exoskeletons of small crustaceans, are eliminated through its mouth. Many flatworms also have a gastrovascular cavity (see Figure 33.10).

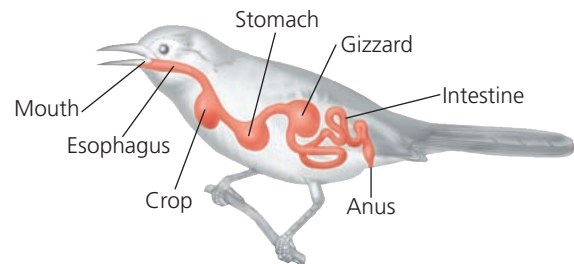
In contrast with cnidarians and flatworms, most animals have a digestive tube extending between two openings, a mouth and an anus (Figure 41.8). Such a tube is called a *complete digestive tract* or, more commonly, an **alimentary canal**. Because food moves along the alimentary canal in a



(a) **Earthworm.** The alimentary canal of an earthworm includes a muscular pharynx that sucks food in through the mouth. Food passes through the esophagus and is stored and moistened in the crop. Mechanical digestion occurs in the muscular gizzard, which pulverizes food with the aid of small bits of sand and gravel. Further digestion and absorption occur in the intestine.



(b) **Grasshopper.** A grasshopper has several digestive chambers grouped into three main regions: a foregut, with an esophagus and crop; a midgut; and a hindgut. Food is moistened and stored in the crop, but most digestion occurs in the midgut. Pouches called gastric caeca (singular, caeca) extend from the beginning of the midgut and function in digestion and absorption.



(c) **Bird.** Many birds have a crop for storing food and a stomach and gizzard for mechanically digesting it. Chemical digestion and absorption of nutrients occur in the intestine.

▲ **Figure 41.8 Alimentary canals.** These examples illustrate variation in the organization and structure of compartments that carry out stepwise digestion, storage, and absorption in different animals.

single direction, the tube can be organized into specialized compartments that carry out digestion and nutrient absorption in a stepwise fashion. An animal with an alimentary canal can ingest food while earlier meals are still being digested, a feat that is likely to be difficult or inefficient for an animal with a gastrovascular cavity. In the next section, we'll explore the organization of a mammalian alimentary canal.

CONCEPT CHECK 41.2

1. Distinguish the overall structure of a gastrovascular cavity from that of an alimentary canal.
2. In what sense are nutrients from a recently ingested meal not really "inside" your body prior to the absorption stage of food processing?
3. **WHAT IF?** Thinking in broad terms, what similarities can you identify between digestion in an animal body and the breakdown of gasoline in an automobile? (You don't have to know about auto mechanics.)

For suggested answers, see Appendix A.

CONCEPT 41.3

Organs specialized for sequential stages of food processing form the mammalian digestive system

Because most animals, including mammals, have an alimentary canal, the mammalian digestive system can serve to illustrate the general principles of food processing. In

mammals, the digestive system consists of the alimentary canal and various accessory glands that secrete digestive juices through ducts into the canal (**Figure 41.9**). The accessory glands of the mammalian digestive system are three pairs of salivary glands, the pancreas, the liver, and the gallbladder.

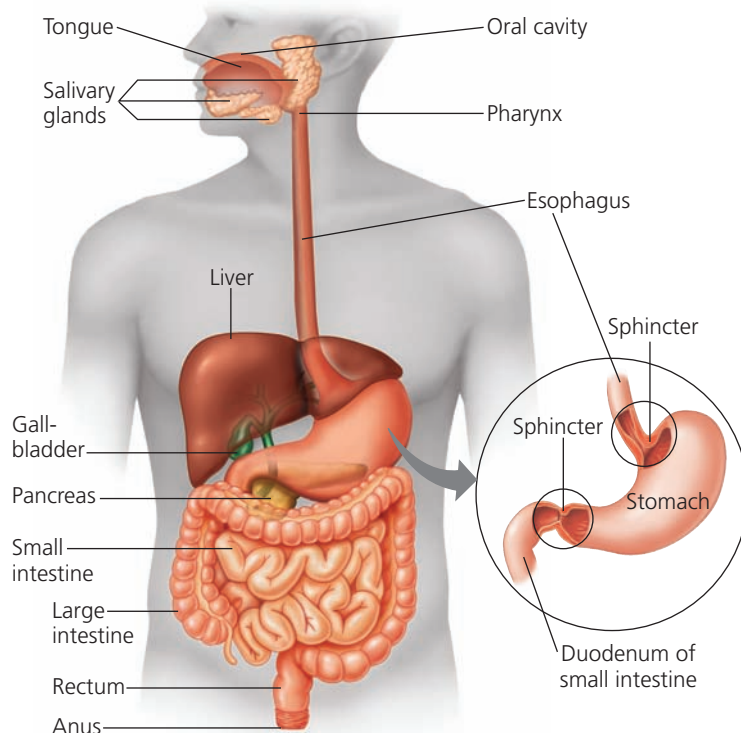
Food is pushed along the alimentary canal by **peristalsis**, alternating waves of contraction and relaxation in the smooth muscles lining the canal. At some of the junctions between specialized compartments, the muscular layer forms ringlike valves called **sphincters**. Acting like drawstrings to close off the alimentary canal, sphincters regulate the passage of material between compartments.

Using the human digestive system as a model, let's now follow a meal through the alimentary canal. As we do so, we'll examine in more detail what happens to the food in each digestive compartment along the way.

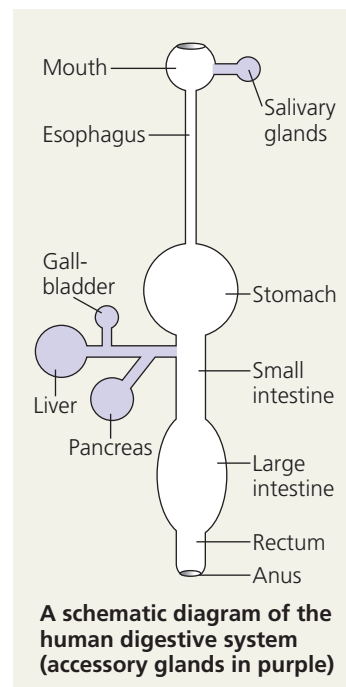
The Oral Cavity, Pharynx, and Esophagus

Ingestion and the initial steps of digestion occur in the mouth, or **oral cavity**. Mechanical digestion begins as teeth of various shapes cut, mash, and grind food, making the food easier to swallow and increasing its surface area. Meanwhile, the **salivary glands** deliver saliva through ducts to the oral cavity. The release of saliva when food enters the mouth is a reflex, an automatic reaction mediated by the nervous system. Saliva may also be released before food enters the mouth, triggered by a learned association between eating and the time of day, a cooking odor, or another stimulus.

► **Figure 41.9 The human digestive system.** After food is chewed and swallowed, it takes 5–10 seconds for it to pass down the esophagus and into the stomach, where it spends 2–6 hours being partially digested. Final digestion and nutrient absorption occur in the small intestine over a period of 5–6 hours. Within 12–24 hours, any undigested material passes through the large intestine, and feces are expelled through the anus.



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Saliva initiates chemical digestion while also protecting the oral cavity. The enzyme **amylase**, found in saliva, hydrolyzes starch (a glucose polymer from plants) and glycogen (a glucose polymer from animals) into smaller polysaccharides and the disaccharide maltose. Much of the protective effect of saliva is provided by **mucus**, a viscous mixture of water, salts, cells, and slippery glycoproteins (carbohydrate-protein complexes) called mucins. Mucus in saliva protects the lining of the mouth from abrasion and lubricates food for easier swallowing. Additional components of saliva include buffers, which help prevent tooth decay by neutralizing acid, and antimicrobial agents (such as lysozyme; see Figure 5.16), which protect against bacteria that enter the mouth with food.

Much as a doorman screens and assists people entering a fancy hotel, the tongue aids digestive processes by evaluating ingested material and then enabling its further passage. When food arrives at the oral cavity, the tongue plays a critical role in distinguishing which foods should be processed further. (See Chapter 50 for a discussion of the sense of taste.) After food is deemed acceptable and chewing commences, tongue movements manipulate the mixture of saliva and food, helping shape it into a ball called a **bolus**. During swallowing, the tongue provides further help, pushing the bolus to the back of the oral cavity and into the pharynx.

The **pharynx**, or throat region, opens to two passageways: the trachea (windpipe) and the esophagus (**Figure 41.10**). The trachea leads to the lungs (see Figure 42.23), whereas the **esophagus** connects to the stomach. Once food enters the esophagus, peristaltic contractions of smooth muscle move each bolus to the stomach.

Swallowing must be carefully choreographed to keep food and liquids from entering the trachea and causing choking, a blockage of the trachea. The resulting lack of airflow into the lungs can be fatal if the material is not dislodged by

vigorous coughing, a series of back slaps, or a forced upward thrust of the diaphragm (the Heimlich maneuver).

Digestion in the Stomach

The **stomach**, which is located just below the diaphragm, stores food and begins digestion of proteins. With accordion-like folds and a very elastic wall, this organ can stretch to accommodate about 2 L of food and fluid. The stomach secretes a digestive fluid called **gastric juice** and mixes it with the food through a churning action. This mixture of ingested food and gastric juice is called **chyme**.

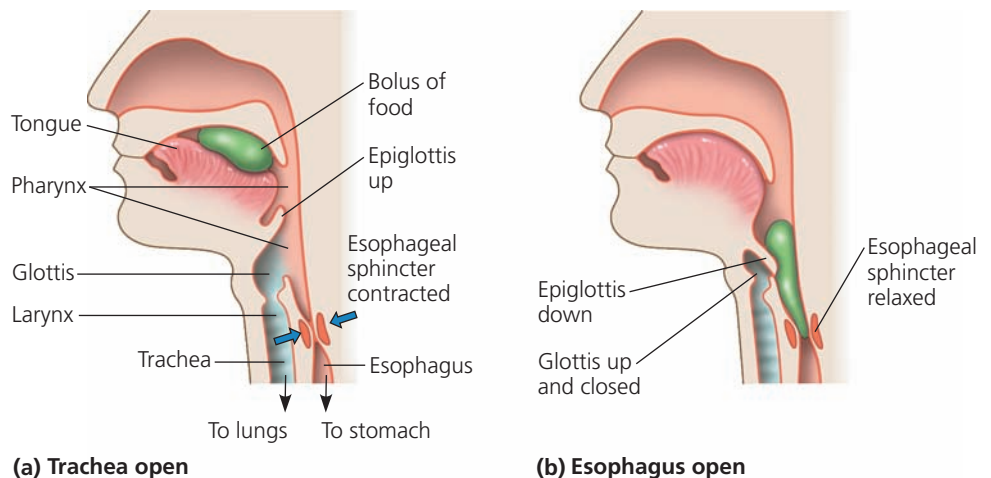
Chemical Digestion in the Stomach

Two components of gastric juice carry out chemical digestion. One is hydrochloric acid (HCl), which disrupts the extracellular matrix that binds cells together in meat and plant material. The concentration of HCl is so high that the pH of gastric juice is about 2, acidic enough to dissolve iron nails (and to kill most bacteria). This low pH denatures (unfolds) proteins in food, increasing exposure of their peptide bonds. The exposed bonds are attacked by the second component of gastric juice—a **protease**, or protein-digesting enzyme, called **pepsin**. Unlike most enzymes, pepsin works best in a very acidic environment. By breaking peptide bonds, it cleaves proteins into smaller polypeptides. Further digestion to individual amino acids occurs in the small intestine.

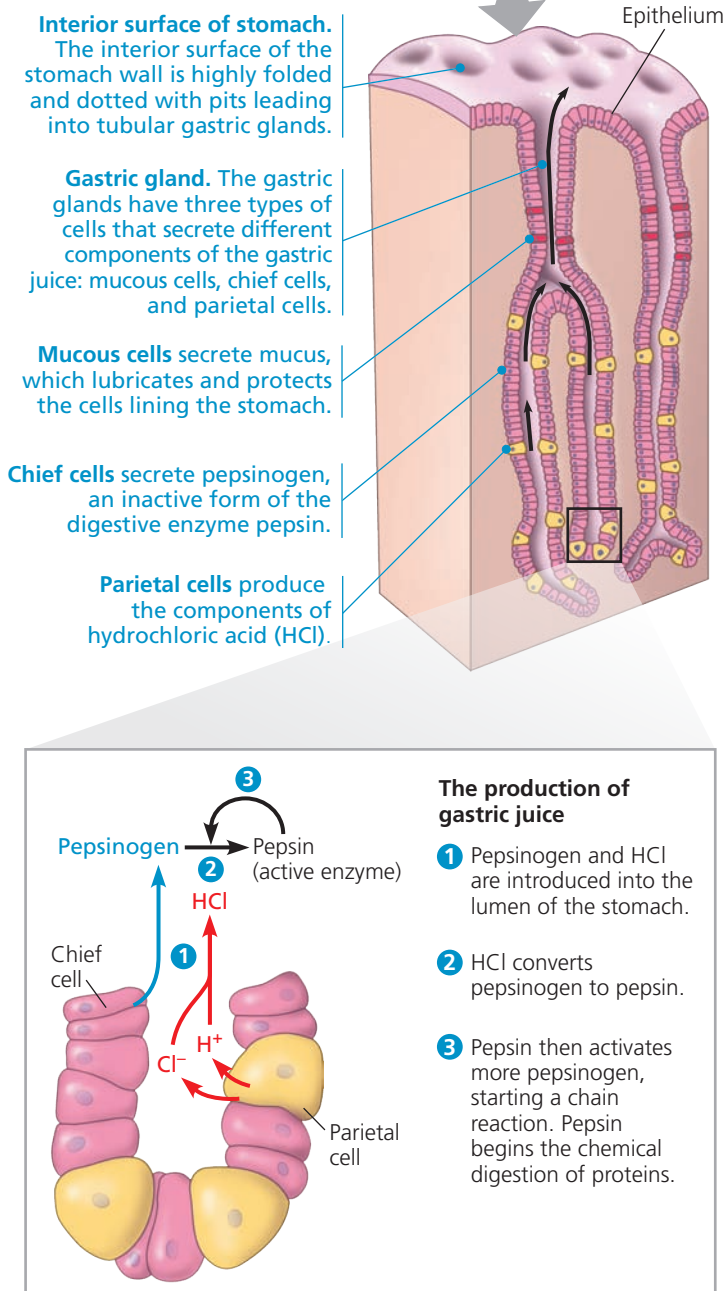
Why doesn't gastric juice destroy the stomach cells that make it? The answer is that the ingredients of gastric juice are kept inactive until they are released into the lumen (cavity) of the stomach.

The components of gastric juice are produced by two types of cells in the gastric glands of the stomach. *Parietal cells* use an ATP-driven pump to expel hydrogen ions into the lumen. At the same time, chloride ions diffuse into the

► **Figure 41.10** Intersection of the human airway and digestive tract. In humans, the pharynx connects to the trachea and the esophagus. **(a)** At most times, a contracted sphincter seals off the esophagus while the trachea remains open. **(b)** When a food bolus arrives at the pharynx, the swallowing reflex is triggered. Movement of the larynx, the upper part of the airway, tips a flap of tissue called the epiglottis down, preventing food from entering the trachea. At the same time, the esophageal sphincter relaxes, allowing the bolus to pass into the esophagus. The trachea then reopens, and peristaltic contractions of the esophagus move the bolus to the stomach.



lumen through specific membrane channels of the parietal cells. It is therefore only within the lumen that hydrogen and chloride ions combine to form HCl (**Figure 41.11**). Meanwhile, *chief cells* release pepsin into the lumen in an inactive form called **pepsinogen**. HCl converts pepsinogen to active pepsin by clipping off a small portion of the molecule and exposing its active site. Through these processes, both HCl and pepsin form in the lumen of the stomach, not within the cells of the gastric glands.



▲ **Figure 41.11** The stomach and its secretions.

After hydrochloric acid converts a small amount of pepsinogen to pepsin, pepsin itself helps activate the remaining pepsinogen. Pepsin, like HCl, can clip pepsinogen to expose the enzyme's active site. This generates more pepsin, which activates more pepsinogen. This series of events is an example of positive feedback (see Concept 40.2).

Why don't HCl and pepsin eat through the lining of the stomach? For one thing, mucus secreted by cells in gastric glands protects against self-digestion (see Figure 41.11). In addition, cell division adds a new epithelial layer every three days, replacing cells before they are fully eroded by digestive juices. Under certain circumstances, however, damaged areas of the stomach lining called gastric ulcers can appear. It had been thought that they were caused by psychological stress and resulting excess acid secretion. However, Australian researchers Barry Marshall and Robin Warren discovered that infection by the acid-tolerant bacterium *Helicobacter pylori* causes ulcers. They also demonstrated that an antibiotic could cure most gastric ulcers. For these findings, they were awarded the Nobel Prize in 2005.

Stomach Dynamics

Chemical digestion by gastric juice is facilitated by the churning action of the stomach. This coordinated series of muscle contractions and relaxations mixes the stomach contents about every 20 seconds. As a result of mixing and enzyme action, what begins as a recently swallowed meal becomes the acidic, nutrient-rich broth known as chyme. Most of the time, sphincters close off the stomach at both ends (see Figure 41.9). The sphincter between the esophagus and the stomach normally opens only when a bolus arrives. Occasionally, however, a person experiences acid reflux, a backflow of chyme from the stomach into the lower end of the esophagus. The resulting irritation of the esophagus is commonly called "heartburn."

Peristaltic contractions typically empty the contents of the stomach into the small intestine within 2–6 hours after a meal. The sphincter located where the stomach opens to the small intestine helps regulate passage into the small intestine, allowing only one squirt of chyme at a time.

Digestion in the Small Intestine

Although chemical digestion of some nutrients begins in the oral cavity or stomach, most enzymatic hydrolysis of the macromolecules from food occurs in the small intestine (**Figure 41.12**). The **small intestine** is the alimentary canal's longest compartment—over 6 m (20 feet) long in humans! Its name refers to its small diameter, compared with that of the large intestine. The first 25 cm (10 inches) or so of the small intestine forms the **duodenum**. It is here that chyme from the stomach mixes with digestive juices from the pancreas, liver, and gallbladder, as well as from gland cells of the

intestinal wall itself. As you will see in Concept 41.5, hormones released by the stomach and duodenum control the digestive secretions into the alimentary canal.

Pancreatic Secretions

The **pancreas** aids chemical digestion by producing an alkaline solution rich in bicarbonate as well as several enzymes (see Figure 41.12). The bicarbonate neutralizes the acidity of chyme and acts as a buffer. Among the pancreatic enzymes are trypsin and chymotrypsin, proteases secreted into the duodenum in inactive forms. In a chain reaction similar to the activation of pepsin, they are activated when safely located in the lumen of the duodenum.

Bile Production by the Liver

Digestion of fats and other lipids begins in the small intestine and relies on the production of **bile**, a mixture of substances

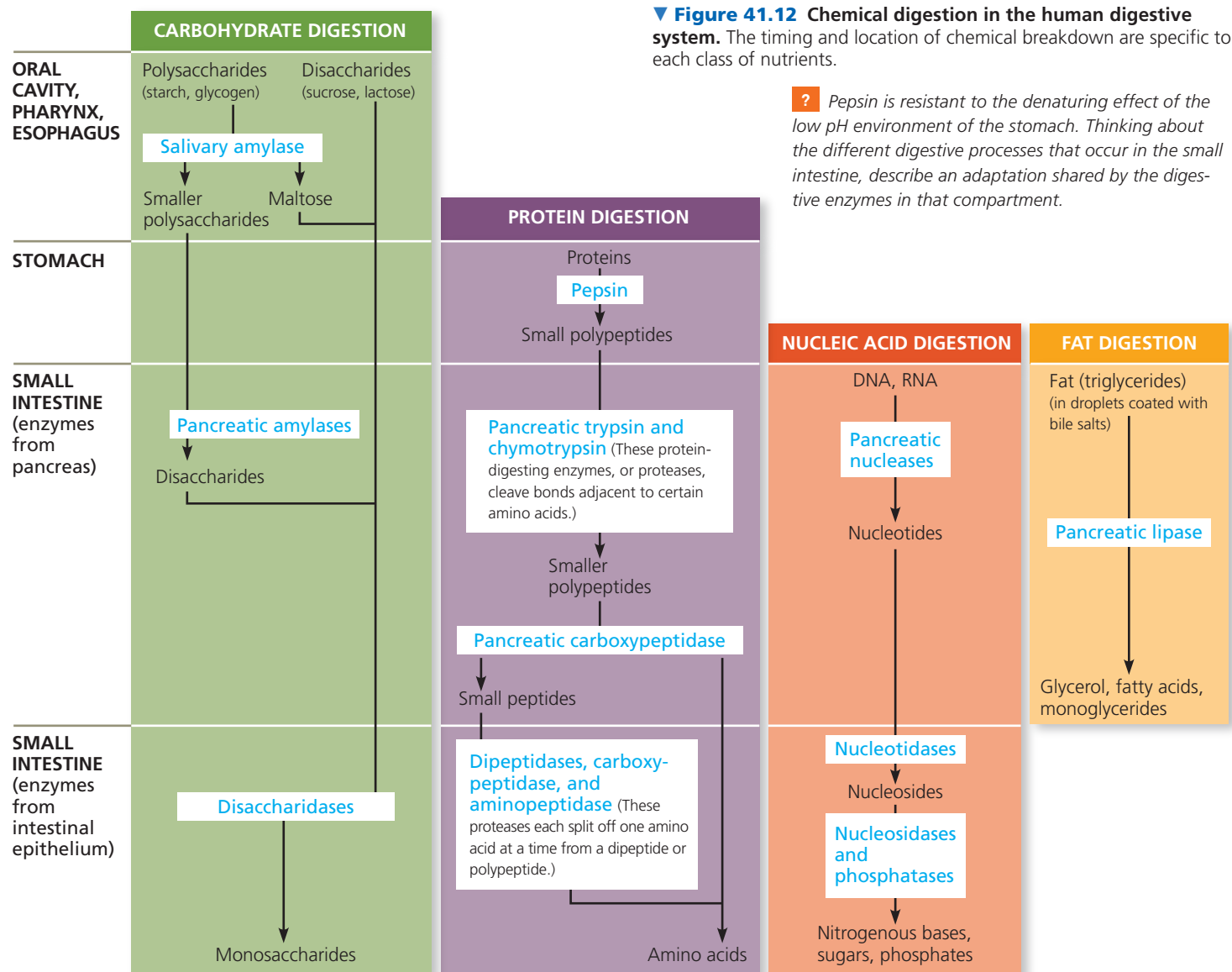
that is made in the **liver**. Bile contains bile salts, which act as emulsifiers (detergents) that aid in digestion and absorption of lipids. Bile is stored and concentrated in the **gallbladder**.

Bile production is integral to one of the other vital functions of the liver: the destruction of red blood cells that are no longer fully functional. In producing bile, the liver incorporates some pigments that are by-products of red blood cell disassembly. These bile pigments are then eliminated from the body with the feces. In some liver or blood disorders, bile pigments accumulate in the skin, resulting in a characteristic yellowing called jaundice.

Secretions of the Small Intestine

The epithelial lining of the duodenum is the source of several digestive enzymes (see Figure 41.12). Some are secreted into the lumen of the duodenum, whereas others are bound to the surface of epithelial cells.

▼ **Figure 41.12** Chemical digestion in the human digestive system. The timing and location of chemical breakdown are specific to each class of nutrients.



? Pepsin is resistant to the denaturing effect of the low pH environment of the stomach. Thinking about the different digestive processes that occur in the small intestine, describe an adaptation shared by the digestive enzymes in that compartment.

While enzymatic hydrolysis proceeds, peristalsis moves the mixture of chyme and digestive juices along the small intestine. Most digestion is completed in the duodenum. The remaining regions of the small intestine, called the *jejunum* and *ileum*, are the major sites for absorption of nutrients, as discussed next.

Absorption in the Small Intestine

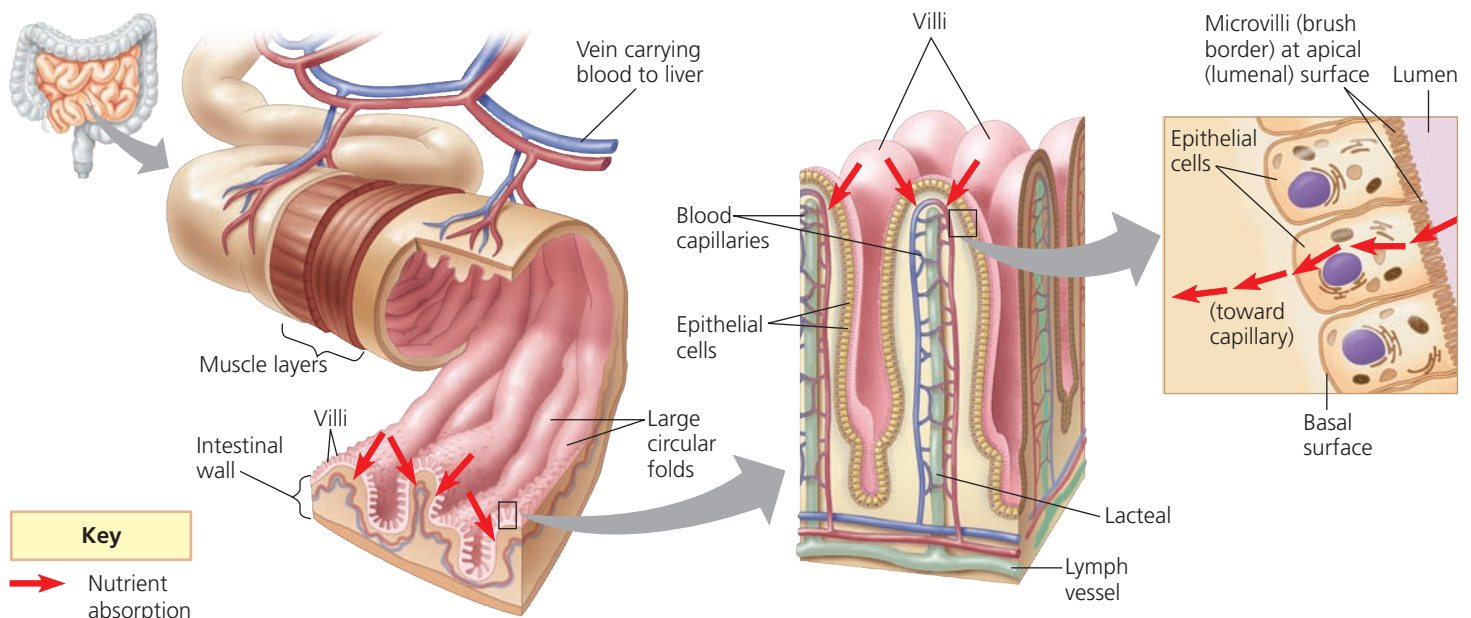
To reach body tissues, nutrients in the lumen must first be absorbed across the lining of the alimentary canal. Most of this absorption occurs at the highly folded surface of the small intestine, as illustrated in **Figure 41.13**. Large folds in the lining encircle the intestine and are studded with finger-like projections called **villi**. In turn, each epithelial cell of a villus has on its apical surface many microscopic projections, or **microvilli**, that are exposed to the intestinal lumen. The many side-by-side microvilli give cells of the intestinal epithelium a brush-like appearance that is reflected in the name *brush border*. Together, the folds, villi, and microvilli of the small intestine have a surface area of 200–300 m², roughly the size of a tennis court. This enormous surface area is an evolutionary adaptation that greatly increases the rate of nutrient absorption (see Figure 33.9 for more discussion and examples of maximizing surface area in diverse organisms).

Depending on the nutrient, transport across the epithelial cells can be passive or active (see Chapter 7). The sugar fructose, for example, moves by facilitated diffusion down its concentration gradient from the lumen of the small

intestine into the epithelial cells. From there, fructose exits the basal surface and is absorbed into microscopic blood vessels, or capillaries, at the core of each villus. Other nutrients, including amino acids, small peptides, vitamins, and most glucose molecules, are pumped against concentration gradients into the epithelial cells of the villus. This active transport allows much more absorption of those nutrients than would be possible with passive diffusion alone.

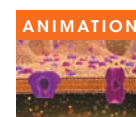
The capillaries and veins that carry nutrient-rich blood away from the villi converge into the **hepatic portal vein**, a blood vessel that leads directly to the liver. From the liver, blood travels to the heart and then to other tissues and organs. This arrangement serves two major functions. First, it allows the liver to regulate the distribution of nutrients to the rest of the body. Because the liver can interconvert many organic molecules, blood that leaves the liver may have a very different nutrient balance than the blood that entered. Second, the arrangement allows the liver to remove toxic substances before the blood circulates broadly. The liver is the primary site for the detoxification of many organic molecules, including drugs, that are foreign to the body.

Although many nutrients leave the small intestine through the bloodstream, some products of fat (triglyceride, also known as triacylglycerol) digestion take a different path (**Figure 41.14**). Hydrolysis of fats by lipase in the small intestine generates fatty acids and monoglycerides. (A monoglyceride is a single fatty acid joined to glycerol.) These products are absorbed by epithelial cells and recombined into triglycerides. They are then coated with phospholipids, cholesterol,

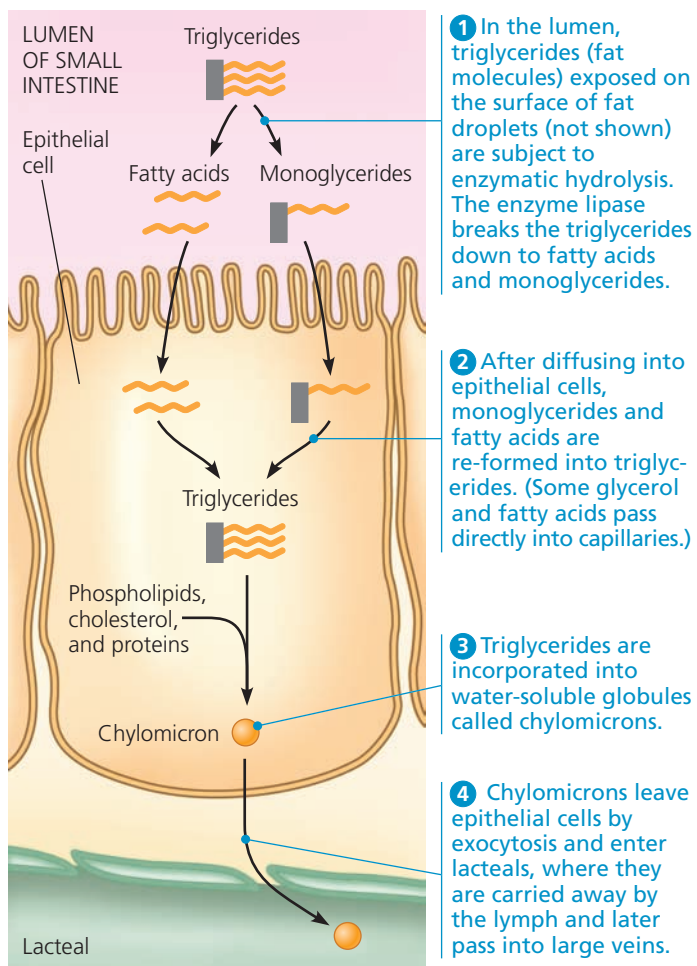


▲ Figure 41.13 Nutrient absorption in the small intestine.

? Tapeworms sometimes infect the human alimentary canal, anchoring themselves to the wall of the small intestine. Based on how digestion is compartmentalized along the mammalian alimentary canal, what digestive functions would you expect these parasites to have?



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▲ **Figure 41.14 Absorption of fats.** Because fats are insoluble in water, adaptations are needed to digest and absorb them. Bile salts (not shown) break up large fat droplets and maintain a small droplet size in the intestinal lumen, exposing more of the fat at the surface to enzymatic hydrolysis. The fatty acids and monoglycerides released by hydrolysis can diffuse into epithelial cells, where fats are reassembled and incorporated into water-soluble chylomicrons that enter the lymphatic system.

and proteins, forming globules called **chylomicrons**. Being water soluble, chylomicrons can dissolve in the blood and travel via the circulatory system.

Before reaching the bloodstream, chylomicrons are first transported from an epithelial cell in the intestine into a **lacteal**, a vessel at the core of each villus (see Figures 41.13 and 41.14). Lacteals are part of the vertebrate lymphatic system, which is a network of vessels filled with a clear fluid called lymph. Starting at the lacteals, lymph containing the chylomicrons passes into the larger vessels of the lymphatic system and eventually into large veins that return the blood to the heart.

In addition to absorbing nutrients, the small intestine has an important function in the recovery of water and ions. Each day we consume about 2 L of water and secrete another 7 L in digestive juices. Typically all but 0.1 L of the water is reabsorbed in the intestines, with most of the

recovery occurring in the small intestine. There is no mechanism for active transport of water. Instead, water is reabsorbed by osmosis when sodium and other ions are pumped out of the lumen of the intestine.

Processing in the Large Intestine

The alimentary canal ends with the **large intestine**, which includes the colon, cecum, and rectum. The small intestine connects to the large intestine at a T-shaped junction (**Figure 41.15**). One arm of the T is the 1.5-m-long **colon**, which leads to the rectum and anus. The other arm is a pouch called the **cecum**. The cecum is important for fermenting ingested material, especially in animals that eat large amounts of plant material.

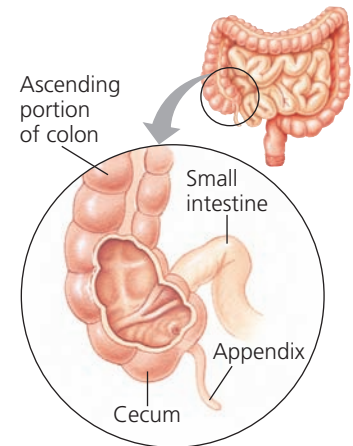
Compared with many other mammals, humans have a small cecum. The **appendix**, a finger-like extension of the human cecum, has a minor and dispensable role in immunity.

The colon completes the reabsorption of water that began in the small intestine. What remain are the **feces**, the wastes of the digestive system, which become increasingly solid as they are moved along the colon by peristalsis. It takes approximately 12–24 hours for material to travel the length of the colon. If the lining of the colon is irritated—by a viral or bacterial infection, for instance—less water than normal may be reabsorbed, resulting in diarrhea. The opposite problem, constipation, occurs when the feces move along the colon too slowly. Too much water is reabsorbed, and the feces become compacted.

The undigested material in feces includes cellulose fiber. Although it provides no caloric value (energy) to humans, fiber helps move food along the alimentary canal.

A rich community of mostly harmless bacteria lives on the unabsorbed organic material in the human colon, contributing approximately one-third of the dry weight of feces. As by-products of their metabolism, many colon bacteria generate gases, including methane and hydrogen sulfide, the latter of which has an offensive odor. These gases and ingested air are expelled through the anus.

The terminal portion of the large intestine is the **rectum**, where the feces are stored until they can be eliminated. Between the rectum and the anus are two sphincters, the inner one being involuntary and the outer one being voluntary. Periodically, strong contractions of the colon create an urge



▲ **Figure 41.15 Junction of the small and large intestines.**

to defecate. Because filling of the stomach triggers a reflex that increases the rate of contractions in the colon, the urge to defecate often follows a meal.

We have followed a meal from one opening (the mouth) of the alimentary canal to the other (the anus). Next we'll look at some adaptations of this general digestive plan in different animals.

CONCEPT CHECK 41.3

1. Explain why a proton pump inhibitor, such as the drug Prilosec, relieves the symptoms of acid reflux.
2. Thinking about our nutritional needs and feeding behavior, propose an evolutionary explanation for why amylase, unlike other digestive enzymes, is secreted into the mouth.
3. **WHAT IF?** If you mixed gastric juice with crushed food in a test tube, what would happen?

For suggested answers, see Appendix A.

CONCEPT 41.4

Evolutionary adaptations of vertebrate digestive systems correlate with diet

EVOLUTION The digestive systems of mammals and other vertebrates are variations on a common plan, but there are many intriguing adaptations, often associated with the animal's diet. To highlight how form fits function, we'll examine a few of them.

Dental Adaptations

Dentition, an animal's assortment of teeth, is one example of structural variation reflecting diet (**Figure 41.16**). The

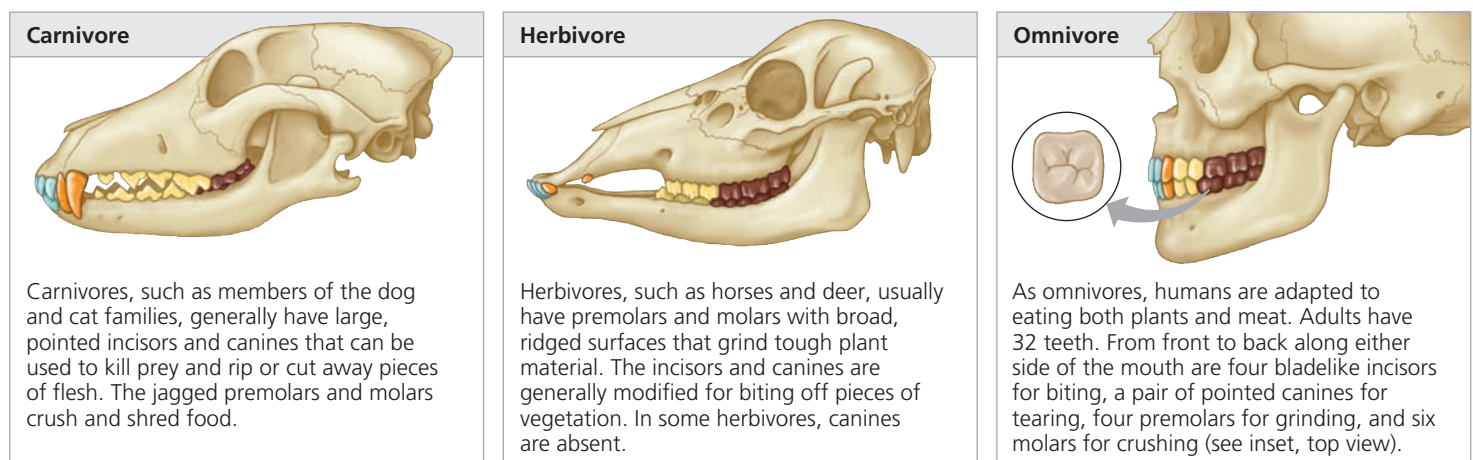
evolutionary adaptation of teeth for processing different kinds of food is one of the major reasons mammals have been so successful. For example, the sea otter in **Figure 41.1** uses its sharp canine teeth to tear apart prey such as crabs and its slightly rounded molars to crush their shells. Nonmammalian vertebrates generally have less specialized dentition, but there are interesting exceptions. Venomous snakes, such as rattlesnakes, have fangs, modified teeth that inject venom into prey. Some fangs are hollow, like syringes, whereas others drip the toxin along grooves on the surfaces of the teeth.

Stomach and Intestinal Adaptations

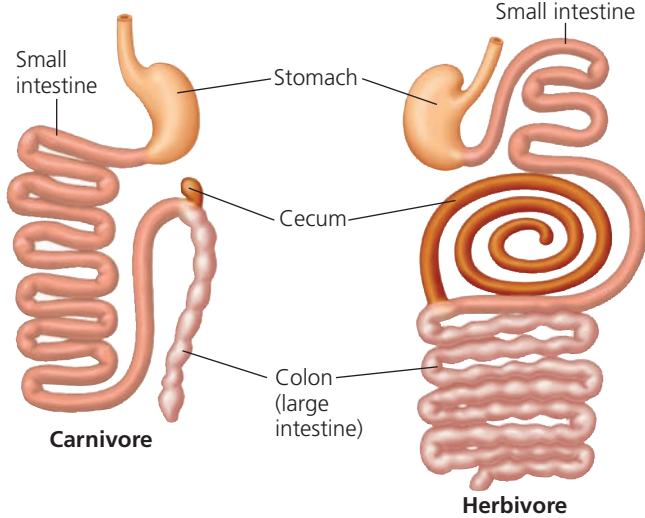
Evolutionary adaptations to differences in diet are sometimes apparent as variations in the dimensions of digestive organs. For example, large, expandable stomachs are common in carnivorous vertebrates, which may wait a long time between meals and must eat as much as they can when they do catch prey. An expandable stomach enables a rock python to ingest a whole gazelle (see **Figure 41.6**) and a 200-kg African lion to consume 40 kg of meat in one meal!

Adaptation is also apparent in the length of the digestive system in different vertebrates. In general, herbivores and omnivores have longer alimentary canals relative to their body size than do carnivores. Plant matter is more difficult to digest than meat because it contains cell walls. A longer digestive tract furnishes more time for digestion and more surface area for the absorption of nutrients. As an example, consider the coyote and koala in **Figure 41.17**. Although these two mammals are about the same size, the koala's intestines are much longer, enhancing the processing of fibrous, protein-poor eucalyptus leaves from which the koala obtains nearly all of its nutrients and water.

▼ **Figure 41.16** Dentition and diet.



Key ■ Incisors ■ Canines ■ Premolars ■ Molars



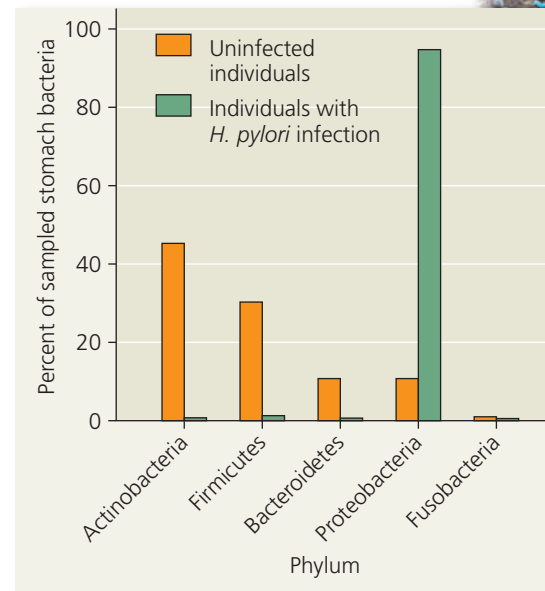
▲ **Figure 41.17** The alimentary canals of a carnivore (coyote) and herbivore (koala). The relatively short digestive tract of the coyote is sufficient for digesting meat and absorbing its nutrients. In contrast, the koala's long alimentary canal is specialized for digesting eucalyptus leaves. Extensive chewing chops the leaves into tiny pieces, increasing exposure to digestive juices. In the long cecum and the upper portion of the colon, symbiotic bacteria further digest the shredded leaves, releasing nutrients that the koala can absorb.

Mutualistic Adaptations

An estimated 10–100 trillion bacteria live in the human digestive system. One bacterial inhabitant, *Escherichia coli*, is so common in the digestive system that its presence in lakes and streams is a useful indicator of contamination by untreated sewage.

The coexistence of humans and many of these bacteria involves mutualistic symbiosis, a mutually beneficial interaction between two species (see Concept 54.1). For example, some intestinal bacteria produce vitamins, such as vitamin K, biotin, and folic acid, that supplement our dietary intake when absorbed into the blood. Intestinal bacteria also regulate the development of the intestinal epithelium and the function of the innate immune system.

Recently, we have greatly expanded our knowledge of the collection of bacteria, called the *microbiome*, in the human digestive system. To identify these bacteria, both beneficial and harmful, scientists are using a DNA sequencing approach based on the polymerase chain reaction (see Figure 20.8). They have found more than 400 bacterial species in the human digestive tract, a far greater number than had been identified through approaches relying on laboratory culture and characterization.



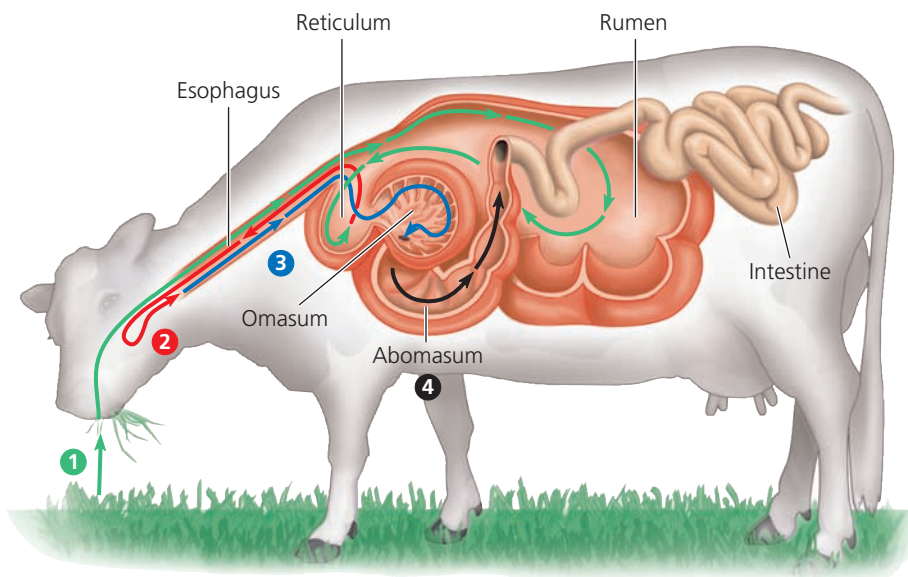
▲ **Figure 41.18** The stomach microbiome. By copying and sequencing bacterial DNA in samples obtained from human stomachs, researchers characterized the bacterial community that makes up the stomach microbiome. In samples from individuals infected with *Helicobacter pylori*, more than 95% of the sequences were from that species, which belongs to the phylum Proteobacteria. The stomach microbiome in uninfected individuals was much more diverse.

One recent microbiome study provided an important clue as to why the bacterium *H. pylori* disrupts stomach health, leading to ulcers. After collecting stomach tissue from uninfected and *H. pylori*-infected adults, researchers identified all the bacterial species in each sample. What they found was remarkable: *H. pylori* infection led to a near complete elimination from the stomach of all other bacterial species (Figure 41.18). Such studies on differences in the microbiome as a result of particular diseases holds promise for the development of new and more effective therapies.

Mutualistic Adaptations in Herbivores

Mutualistic symbiosis is particularly important in herbivores. Much of the chemical energy in herbivore diets comes from the cellulose of plant cell walls, but animals do not produce enzymes that hydrolyze cellulose. Instead, many vertebrates (as well as termites, whose wooden diets consist largely of cellulose) host large populations of mutualistic bacteria and protists in fermentation chambers in their alimentary canals. These microorganisms have enzymes that can digest cellulose to simple sugars and other compounds that the animal can absorb. In many cases, the microorganisms also use the sugars from digested cellulose in the production of a variety of nutrients essential to the animal, such as vitamins and amino acids.

In horses, koalas, and elephants, mutualistic microorganisms are housed in a large cecum. In contrast, the hoatzin, an herbivorous bird found in South American rain forests,



◀ **Figure 41.19 Ruminant digestion.** The stomach of a cow, a ruminant, has four chambers. **1** Chewed food first enters the rumen and reticulum, where mutualistic microorganisms digest cellulose in the plant material. **2** Periodically, the cow regurgitates and rechews “cud” from the reticulum, further breaking down fibers and thereby enhancing microbial action. **3** The reswallowed cud passes to the omasum, where some water is removed. **4** It then passes to the abomasum, for digestion by the cow’s enzymes. In this way, the cow obtains significant nutrients from both the grass and the mutualistic microorganisms, which maintain a stable population in the rumen.

hosts microorganisms in a large, muscular crop (an esophageal pouch; see Figure 41.8). Hard ridges in the wall of the crop grind plant leaves into small fragments, and the microorganisms break down cellulose.

In rabbits and some rodents, mutualistic bacteria live in the large intestine as well as in the cecum. Since most nutrients are absorbed in the small intestine, nourishing by-products of fermentation by bacteria in the large intestine are initially lost with the feces. Rabbits and rodents recover these nutrients by *coprophagy* (from the Greek, meaning “dung eating”), feeding on some of their feces and then passing the food through the alimentary canal a second time. The familiar rabbit “pellets,” which are not reingested, are the feces eliminated after food has passed through the digestive tract twice.

The most elaborate adaptations for an herbivorous diet have evolved in the animals called *ruminants*, the cud-chewing animals that include deer, sheep, and cattle (**Figure 41.19**).

Although we have focused our discussion on vertebrates, adaptations related to digestion are also widespread among other animals. Some of the most remarkable examples are the giant tubeworms (over 3 m long) that live at pressures as high as 260 atmospheres around deep-sea hydrothermal vents (see Figure 52.14). These worms have no mouth or digestive system. Instead, they obtain all of their energy and nutrients from mutualistic bacteria that live within their bodies. The bacteria carry out chemoautotrophy (see



▲ **Giant tubeworm**

Concept 27.3) using the carbon dioxide, oxygen, hydrogen sulfide, and nitrate available at the vents. Thus, for invertebrates and vertebrates alike, mutualistic symbiosis has evolved as an adaptation that expands the sources of nutrition available to animals. Having examined how animals optimize their extraction of nutrients from food, we’ll next turn to the challenge of balancing the use of these nutrients.

CONCEPT CHECK 41.4

1. What are two advantages of a longer alimentary canal for processing plant material that is difficult to digest?
2. What features of a mammal’s digestive system make it an attractive habitat for mutualistic microorganisms?
3. **WHAT IF?** “Lactose-intolerant” people have a shortage of lactase, the enzyme that breaks down lactose in milk. As a result, they sometimes develop cramps, bloating, or diarrhea after consuming dairy products. Suppose such a person ate yogurt containing bacteria that produce lactase. Why would eating yogurt likely provide at best only temporary relief of the symptoms?

For suggested answers, see Appendix A.

CONCEPT 41.5

Feedback circuits regulate digestion, energy storage, and appetite

The processes that enable an animal to obtain nutrients are matched to the organism’s circumstances and need for energy—an example of evolutionary adaptation.

Regulation of Digestion

Many animals have long intervals between meals and do not need their digestive systems to be active continuously.

Instead, each step in processing is activated as food reaches a new compartment in the alimentary canal. The arrival of food triggers the secretion of substances that promote the next stage of chemical digestion, as well as muscular contractions that propel food farther along the canal. For example, you learned earlier that nervous reflexes stimulate the release of saliva when food enters the oral cavity and orchestrate swallowing when a bolus of food reaches the pharynx. Similarly, the arrival of food in the stomach triggers churning and the release of gastric juices. A branch of the nervous system called the *enteric division*, which is dedicated to the digestive organs, regulates these events as well as peristalsis in the small and large intestines.

The endocrine system also plays a critical role in controlling digestion. As described in **Figure 41.20**, a series of hormones released by the stomach and duodenum help ensure that digestive secretions are present only when needed. Like all hormones, they are transported through the bloodstream. This is true even for the hormone gastrin, whose target (the stomach) is the same organ that secretes it.

Regulation of Energy Storage

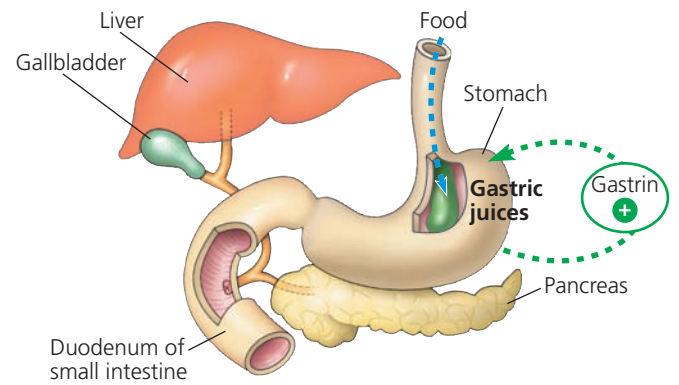
When an animal takes in more energy-rich molecules than it needs for metabolism and activity, it stores the excess energy (see Concept 40.4). In concluding our overview of nutrition, we'll examine some ways in which animals manage their energy allocation.

In humans, the first sites used for energy storage are liver and muscle cells. In these cells, excess energy from the diet is stored in glycogen, a polymer made up of many glucose units (see Figure 5.6b). Once glycogen depots are full, any additional excess energy is usually stored in fat in adipose cells.

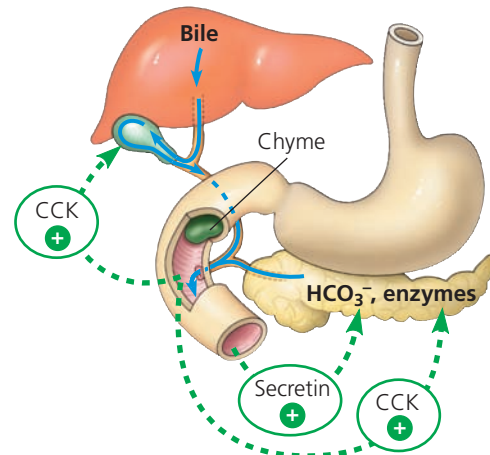
When fewer calories are taken in than are expended—perhaps because of sustained heavy exercise or lack of food—the human body generally expends liver glycogen first and then draws on muscle glycogen and fat. Fats are especially rich in energy; oxidizing a gram of fat liberates about twice the energy liberated from a gram of carbohydrate or protein. For this reason, adipose tissue provides the most space-efficient way for the body to store large amounts of energy. Most healthy people have enough stored fat to sustain them through several weeks without food.

Glucose Homeostasis

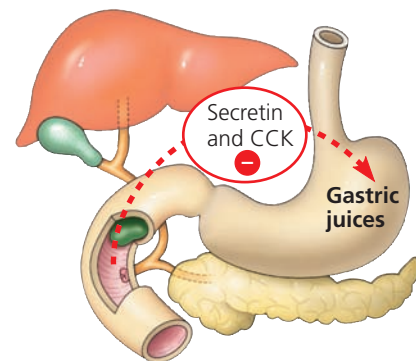
The synthesis and breakdown of glycogen are central not only to energy storage, but also to maintaining metabolic balance through glucose homeostasis. In humans, the normal range for the concentration of glucose in the blood is 70–110 mg/100 mL. Because glucose is a major fuel for cellular respiration and a key source of carbon skeletons for biosynthesis, maintaining blood glucose concentrations near this normal range is critical.



- 1 As food arrives at the stomach, it stretches the stomach walls, triggering release of the hormone *gastrin*. Gastrin circulates via the bloodstream back to the stomach, where it stimulates production of gastric juices.



- 2 Chyme—an acidic mixture of partially digested food—eventually passes from the stomach to the duodenum. The duodenum responds to amino acids or fatty acids in the chyme by releasing the digestive hormones cholecystokinin and secretin. *Cholecystokinin (CCK)* stimulates the release of digestive enzymes from the pancreas and of bile from the gallbladder. *Secretin* stimulates the pancreas to release bicarbonate (HCO_3^-), which neutralizes chyme.



- 3 If the chyme is rich in fats, the high levels of secretin and CCK released act on the stomach to inhibit peristalsis and secretion of gastric juices, thereby slowing digestion.

Key + Stimulation – Inhibition

▲ **Figure 41.20** Hormonal control of digestion.

Glucose homeostasis relies predominantly on the antagonistic (opposing) effects of two hormones, insulin and glucagon (**Figure 41.21**). When the blood glucose level rises above the normal range, the secretion of **insulin** triggers the uptake of glucose from the blood into body cells, decreasing the blood glucose concentration. When the blood glucose level drops below the normal range, the secretion of **glucagon** promotes the release of glucose into the blood from energy stores, such as liver glycogen, increasing the blood glucose concentration.

The liver is a key site for insulin and glucagon action. After a carbohydrate-rich meal, for example, rising levels of insulin promote biosynthesis of glycogen from glucose entering the liver in the hepatic portal vein. Between meals, when blood in the hepatic portal vein has a much lower glucose concentration, glucagon stimulates the liver to break down glycogen, convert amino acids and glycerol to glucose, and release glucose into the blood.

Insulin also acts on nearly all body cells to stimulate glucose uptake from blood. A major exception is brain cells, which can take up glucose whether or not insulin is present. This evolutionary adaptation ensures that the brain almost always has access to circulating fuel, even if supplies are low.

Glucagon and insulin are both produced in the pancreas. Scattered throughout this organ are cell clusters called

pancreatic islets. Each pancreatic islet has *alpha cells*, which make glucagon, and *beta cells*, which make insulin. Like all hormones, insulin and glucagon are secreted into the interstitial fluid and enter the circulatory system.

Overall, hormone-secreting cells make up only 1–2% of the mass of the pancreas. Other cells in the pancreas produce and secrete bicarbonate ions and the digestive enzymes active in the small intestine (see Figure 41.12). These secretions are released into small ducts that empty into the pancreatic duct, which leads to the small intestine. Thus, the pancreas has functions in both the endocrine and digestive systems.

Diabetes Mellitus

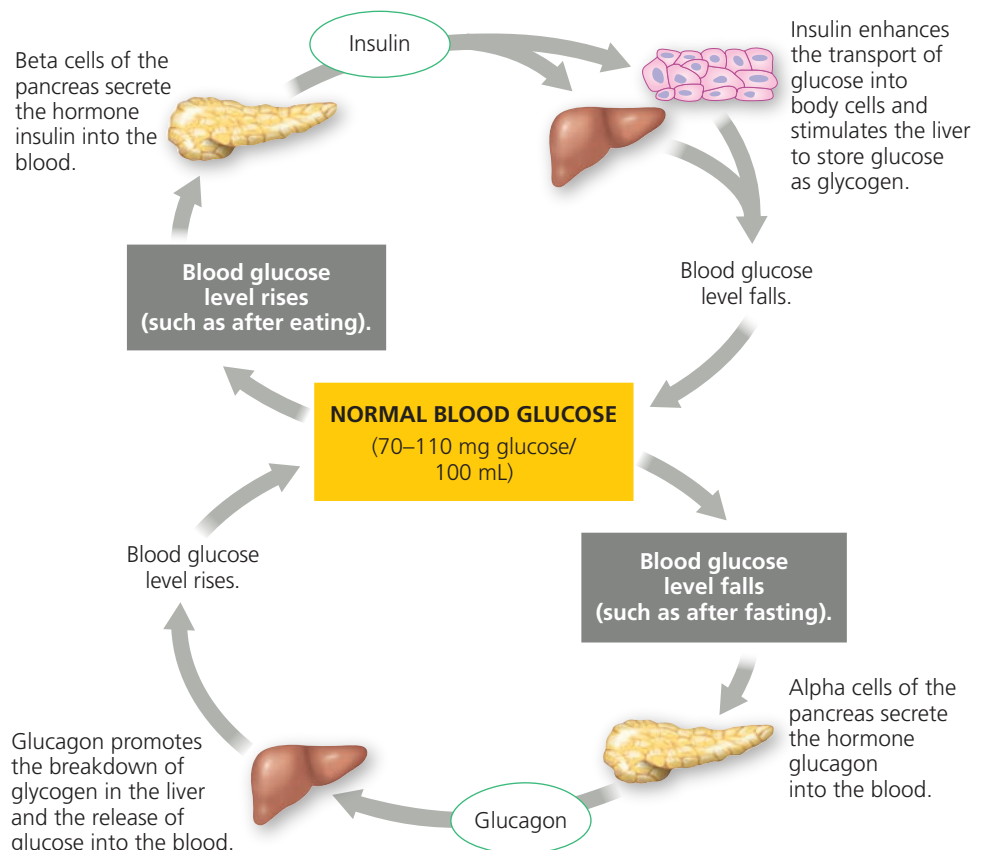
In discussing the role of insulin and glucagon in glucose homeostasis, we have focused exclusively on a healthy metabolic state. However, a number of disorders can disrupt glucose homeostasis with potentially serious consequences, especially for the heart, blood vessels, eyes, and kidneys. The best known and most prevalent of these disorders is diabetes mellitus.

The disease **diabetes mellitus** is caused by a deficiency of insulin or a decreased response to insulin in target tissues. Blood glucose levels rise, but cells are unable to take up enough glucose to meet metabolic needs. Instead, fat becomes the main substrate for cellular respiration. In severe

► **Figure 41.21 Homeostatic regulation of cellular fuel.** After a meal is digested, glucose and other monomers are absorbed into the blood from the digestive tract. The human body regulates the use and storage of glucose, a major cellular fuel.

MAKE CONNECTIONS What form of feedback control does each of these regulatory circuits reflect (see Concept 40.2)?

ANIMATION **BioFlix** Visit the Study Area in **MasteringBiology** for the BioFlix® 3-D Animation on Homeostasis: Regulating Blood Sugar.



cases, acidic metabolites formed during fat breakdown accumulate in the blood, threatening life by lowering blood pH and depleting sodium and potassium ions from the body.

In people with diabetes mellitus, the level of glucose in the blood may exceed the capacity of the kidneys to re-absorb this nutrient. Glucose that remains in the kidney filtrate is excreted. For this reason, the presence of sugar in urine is one test for this disorder. As glucose is concentrated in the urine, more water is excreted along with it, resulting in excessive volumes of urine. *Diabetes* (from the Greek *diabainein*, to pass through) refers to this copious urination, and *mellitus* (from the Greek *meli*, honey) refers to the presence of sugar in urine.

Type 1 Diabetes There are two main types of diabetes mellitus. Each is marked by high blood glucose levels, but with very different causes. *Type 1 diabetes*, or insulin-dependent diabetes, is an autoimmune disorder in which the immune system destroys the beta cells of the pancreas. Type 1 diabetes, which usually appears during childhood, destroys the person's ability to produce insulin. Treatment consists of insulin injections, typically given several times daily. In the past, insulin was extracted from animal pancreases, but now human insulin can be obtained from genetically engineered bacteria, a relatively inexpensive source (see Figure 20.2). Stem cell research may someday provide a cure for type 1 diabetes by generating replacement beta cells that restore insulin production by the pancreas.

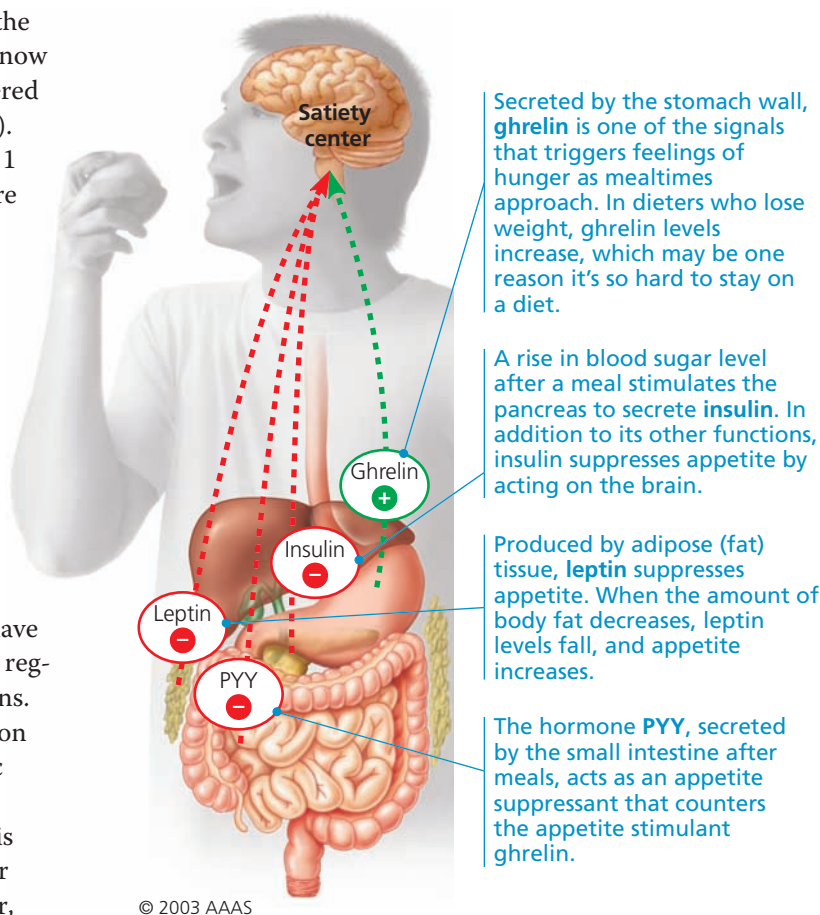
Type 2 Diabetes Non-insulin-dependent diabetes, or *type 2 diabetes*, is characterized by a failure of target cells to respond normally to insulin. Insulin is produced, but target cells fail to take up glucose from the blood, and blood glucose levels remain elevated. Although heredity can play a role in type 2 diabetes, excess body weight and lack of exercise significantly increase the risk. This form of diabetes generally appears after age 40, but even children can develop the disease, particularly if they are overweight and sedentary. More than 90% of people with diabetes have type 2. Many can control their blood glucose levels with regular exercise and a healthy diet; some require medications. Nevertheless, type 2 diabetes is the seventh most common cause of death in the United States and a growing public health problem worldwide.

The resistance to insulin signaling in type 2 diabetes is sometimes due to a genetic defect in the insulin receptor or the insulin response pathway. In many cases, however, events in target cells suppress activity of an otherwise functional response pathway. One source of this suppression appears to be inflammatory signals generated by the innate immune system (see Chapter 43). How obesity and inactivity relate to this suppression is being studied in both humans and laboratory animals.

Regulation of Appetite and Consumption

Consuming more calories than the body needs for normal metabolism, or *overnourishment*, can lead to obesity, the excessive accumulation of fat. Obesity, in turn, contributes to a number of health problems, including type 2 diabetes, cancer of the colon and breast, and cardiovascular disease that can result in heart attacks and strokes. It is estimated that obesity is a factor in about 300,000 deaths per year in the United States alone.

Researchers have discovered several homeostatic mechanisms that operate as feedback circuits controlling the storage and metabolism of fat. A network of neurons relays and integrates information from the digestive system to regulate secretion of hormones that regulate long-term and short-term appetite. The target for these hormones is a "satiety center" in the brain (**Figure 41.22**). For example, *ghrelin*, a hormone secreted by the stomach wall, triggers feelings of hunger before meals. In contrast, both insulin and *PYY*, a hormone secreted by the small intestine after meals,



▲ **Figure 41.22** A few of the appetite-regulating hormones.

Secreted by various organs and tissues, the hormones reach the brain via the bloodstream. These signals act on a region of the brain that in turn controls the "satiety center," which generates the nervous impulses that make us feel either hungry or satiated ("full"). The hormone ghrelin is an appetite stimulant; the other three hormones shown here are appetite suppressants.

SCIENTIFIC SKILLS EXERCISE

Interpreting Data from Experiments with Genetic Mutants

What Are the Roles of the *ob* and *db* Genes in Appetite Regulation? A mutation that disrupts a physiological process is often used to study the normal function of the mutated gene. Ideally, researchers use a standard set of conditions and compare animals that differ genetically only in whether a particular gene is mutant (nonfunctional) or wild-type (normal). In this way, a difference in phenotype, the physiological property being measured, can be attributed to a difference in genotype, the presence or absence of the mutation. To study the role of specific genes in regulating appetite, researchers used laboratory animals with known mutations in those genes.

Mice in which recessive mutations inactivate both copies of either the *ob* gene or the *db* gene eat voraciously and grow much more massive than wild-type mice. In the photograph below, the mouse on the right is wild-type, whereas the obese mouse on the left has an inactivating mutation in both copies of the *ob* gene.



One hypothesis for the normal role of the *ob* and *db* genes is that they participate in a hormone pathway that suppresses appetite when caloric intake is sufficient. Before setting out to isolate the potential hormone, researchers explored this hypothesis genetically.

How the Experiment Was Done The researchers measured the mass of young subject mice of various genotypes and surgically linked the circulatory system of each one to that of another mouse. This procedure ensured that any factor circulating in the bloodstream of either mouse would be transferred to the other in the pair. After eight weeks, they again measured the mass of each subject mouse.

suppress appetite. *Leptin*, a hormone produced by adipose (fat) tissue, also suppresses appetite and appears to play a major role in regulating body fat levels. In the **Scientific Skills Exercise**, you'll interpret data from an experiment studying genes that affect leptin production and function in mice.

Obtaining food, digesting it, and absorbing nutrients are part of the larger story of how animals fuel their activities. Provisioning the body also involves distributing nutrients (circulation), and using nutrients for metabolism requires exchanging respiratory gases with the environment. These processes and the adaptations that facilitate them are the focus of Chapter 42.

Data from the Experiment

	Genotype Pairing (red type indicates mutant genes)		Average Change in Body Mass of Subject (g)
	Subject	Paired with	
(a)	<i>ob</i> ⁺ / <i>ob</i> ⁺ , <i>db</i> ⁺ / <i>db</i> ⁺	<i>ob</i> ⁺ / <i>ob</i> ⁺ , <i>db</i> ⁺ / <i>db</i> ⁺	8.3
(b)	<i>ob</i> / <i>ob</i> , <i>db</i> ⁺ / <i>db</i> ⁺	<i>ob</i> / <i>ob</i> , <i>db</i> ⁺ / <i>db</i> ⁺	38.7
(c)	<i>ob</i> / <i>ob</i> , <i>db</i> ⁺ / <i>db</i> ⁺	<i>ob</i> ⁺ / <i>ob</i> ⁺ , <i>db</i> ⁺ / <i>db</i> ⁺	8.2
(d)	<i>ob</i> / <i>ob</i> , <i>db</i> ⁺ / <i>db</i> ⁺	<i>ob</i> ⁺ / <i>ob</i> ⁺ , <i>db</i> / <i>db</i>	−14.9*

* Due to pronounced weight loss and weakening, subjects in this pairing were remeasured after less than eight weeks.

Interpret the Data

- First, practice reading the genotype information given in the data table. For example, pairing (a) joined two mice that each had the wild-type version of both genes. Describe the two mice in pairing (b), pairing (c), and pairing (d). Explain how each pairing contributed to the experimental design.
- Compare the results observed for pairing (a) and pairing (b) in terms of phenotype. If the results had been identical for these two pairings, what would that outcome have implied about the experimental design?
- Compare the results observed for pairing (c) to those observed for pairing (b). Based on these results, does the *ob*⁺ gene product appear to promote or suppress appetite? Explain your answer.
- Describe the results observed for pairing (d). Note how these results differ from those for pairing (b). Suggest a hypothesis to explain this difference. How could you test your hypothesis using the kinds of mice in this study?

 A version of this Scientific Skills Exercise can be assigned in MasteringBiology.

Data from D. L. Coleman, Effects of parabiosis of obese mice with diabetes and normal mice, *Diabetologia* 9:294–298 (1973).

CONCEPT CHECK 41.5

- Explain how people can become obese even if their intake of dietary fat is relatively low compared with carbohydrate intake.
- WHAT IF?** Suppose you were studying two groups of obese people with genetic abnormalities in the leptin pathway. In one group, the leptin levels are abnormally high; in the other group, they are abnormally low. How would each group's leptin levels change if they ate a low-calorie diet for an extended period? Explain.
- WHAT IF?** An insulinoma is a cancerous mass of pancreatic beta cells that secrete insulin but do not respond to feedback mechanisms. How you would expect an insulinoma to affect blood glucose levels and liver activity?

For suggested answers, see Appendix A.

41 Chapter Review

SUMMARY OF KEY CONCEPTS

- Animals have diverse diets. **Herbivores** mainly eat plants; **carnivores** mainly eat other animals; and **omnivores** eat both. In meeting their nutritional needs, animals must balance consumption, storage, and use of food.

CONCEPT 41.1

An animal's diet must supply chemical energy, organic molecules, and essential nutrients (pp. 893–897)

- Food provides animals with energy for ATP production, carbon skeletons for biosynthesis, and **essential nutrients**—nutrients that must be supplied in preassembled form. Essential nutrients include certain amino acids and fatty acids that animals cannot synthesize; **vitamins**, which are organic molecules; and **minerals**, which are inorganic substances.
- Animals can suffer from two types of malnutrition: an inadequate intake of essential nutrients and a deficiency in sources of chemical energy. Studies of disease at the population level help researchers determine human dietary requirements.

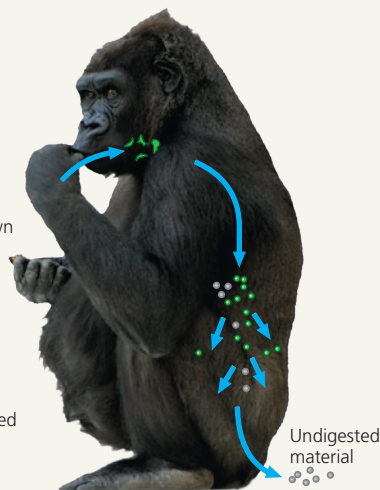
? How can an enzyme cofactor needed for a process that is vital to all animals be an essential nutrient (vitamin) for only some?

CONCEPT 41.2

The main stages of food processing are ingestion, digestion, absorption, and elimination (pp. 897–900)

Stages of food processing

- 1 INGESTION**
(eating)
- 2 DIGESTION**
(enzymatic breakdown of large molecules)
- 3 ABSORPTION**
(uptake of nutrients by cells)
- 4 ELIMINATION**
(passage of undigested materials out of the body in feces)

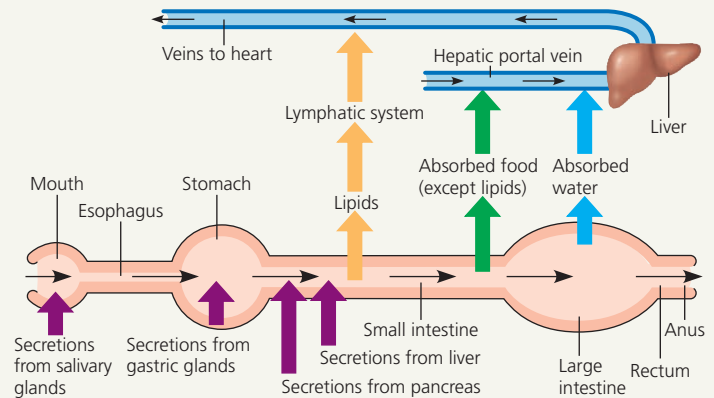


- Animals differ in the ways they obtain and ingest food. Many animals are **bulk feeders**, eating large pieces of food. Other strategies include filter feeding, suspension feeding, and fluid feeding.
- Compartmentalization is necessary to avoid self-digestion. In intracellular digestion, food particles are engulfed by endocytosis and digested within food vacuoles that have fused with lysosomes. In extracellular digestion, which is used by most animals, enzymatic hydrolysis occurs outside cells in a **gastrovascular cavity** or **alimentary canal**.

? Propose an artificial diet that would eliminate the need for one of the first three steps in food processing.

CONCEPT 41.3

Organs specialized for sequential stages of food processing form the mammalian digestive system (pp. 900–906)



? What structural feature of the small intestine makes it better suited for absorption of nutrients than the stomach?

CONCEPT 41.4

Evolutionary adaptations of vertebrate digestive systems correlate with diet (pp. 906–908)

- Vertebrate digestive systems display many evolutionary adaptations associated with diet. For example, dentition, which is the assortment of teeth, generally correlates with diet. In a form of mutualism, many herbivores, including cows, have fermentation chambers where microorganisms digest cellulose. Herbivores also usually have longer alimentary canals than carnivores, reflecting the longer time needed to digest vegetation.

? How does human anatomy indicate that our primate ancestors were not strict vegetarians?

CONCEPT 41.5

Feedback circuits regulate digestion, energy storage, and appetite (pp. 908–912)

- Nutrition** is regulated at multiple levels. Food in the alimentary canal triggers nervous and hormonal responses that control the secretion of digestive juices and that promote the movement of ingested material through the canal. The availability of glucose for energy production is regulated by the hormones **insulin** and **glucagon**, which control the synthesis and breakdown of glycogen.
- Vertebrates store excess calories in glycogen (in liver and muscle cells) and in fat (in adipose cells). These energy stores can be tapped when an animal expends more calories than it consumes. If, however, an animal consumes more calories than it needs for normal metabolism, the resulting overnourishment can lead to the serious health problem of obesity.
- Several hormones, including leptin and insulin, regulate appetite by affecting the brain's satiety center.

? Explain why your stomach might make growling noises when you skip a meal.

TEST YOUR UNDERSTANDING

LEVEL 1: KNOWLEDGE/COMPREHENSION

1. Fat digestion yields fatty acids and glycerol, whereas protein digestion yields amino acids. Both digestive processes
 - a. occur inside cells in most animals.
 - b. add a water molecule to break bonds.
 - c. require a low pH resulting from HCl production.
 - d. consume ATP.
2. The mammalian trachea and esophagus both connect to the
 - a. pharynx.
 - b. stomach.
 - c. large intestine.
 - d. rectum.
3. Which of the following organs is *incorrectly* paired with its function?
 - a. stomach—protein digestion
 - b. large intestine—bile production
 - c. small intestine—nutrient absorption
 - d. pancreas—enzyme production
4. Which of the following is *not* a major activity of the stomach?
 - a. mechanical digestion
 - b. HCl production
 - c. nutrient absorption
 - d. enzyme secretion

LEVEL 2: APPLICATION/ANALYSIS

5. After surgical removal of an infected gallbladder, a person must be especially careful to restrict dietary intake of
 - a. starch.
 - b. protein.
 - c. sugar.
 - d. fat.
6. If you were to jog 1 km a few hours after lunch, which stored fuel would you probably tap?
 - a. muscle proteins
 - b. muscle and liver glycogen
 - c. fat in the liver
 - d. fat in adipose tissue

LEVEL 3: SYNTHESIS/EVALUATION

7. **DRAW IT** Make a flowchart of the events that occur after partially digested food leaves the stomach. Use the following terms: bicarbonate secretion, circulation, decrease in acidity, increase in acidity, secretin secretion, signal detection. Next to each term, indicate the compartment(s) involved. You may use terms more than once.
8. **EVOLUTION CONNECTION**

The human esophagus and trachea share a passage leading from the mouth and nasal passages, which can cause problems. After reviewing vertebrate evolution (see Chapter 34), explain how the evolutionary concept of descent with modification explains this “imperfect” anatomy.

9. SCIENTIFIC INQUIRY

In human populations of northern European origin, the disorder called hemochromatosis causes excess iron uptake from food and affects one in 200 adults. Among adults, men are ten times as likely as women to suffer from iron overload. Taking into account the existence of a menstrual cycle in humans, devise a hypothesis that explains this difference.

10. WRITE ABOUT A THEME: ORGANIZATION

Hair is largely made up of the protein keratin. In a short essay (100–150 words), explain why a shampoo containing protein is not effective in replacing the protein in damaged hair.

11. SYNTHESIZE YOUR KNOWLEDGE



Hummingbirds are well adapted to obtain sugary nectar from flowers, but they use some of the energy obtained from nectar when they forage for insects and spiders. Explain why this foraging is necessary.

For selected answers, see Appendix A.

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